

Round 3 Velvetfruit and VEV Voucher EDA Synthesis

Prosperity 4 Trading Research

April 24, 2026

Scope And Source Files

This document synthesizes two exploratory analyses:

- `phase2/round3/algo/EDA.ipynb`, the first-pass dataset and order-book structure analysis.
- `phase2/round3/algo/ve_vev_eda.py`, the longer Velvetfruit and VEV voucher EDA that generated `ve_vev_eda_output/charts` and `ve_vev_eda_output/tables`.

The products in scope are `VELVETFRUIT_EXTRACT` and the ten call-like voucher products `VEV_4000`, `VEV_4500`, `VEV_5000`, `VEV_5100`, `VEV_5200`, `VEV_5300`, `VEV_5400`, `VEV_5500`, `VEV_6000`, and `VEV_6500`. The full Round 3 data also contains `HYDROGEL_PACK`; both EDAs either filter it out or explicitly skip it for the option-complex analysis.

Executive Findings

- The full combined price dataset has 360,000 rows: 12 products, 30,000 book snapshots each, split into 3 historical days and 10,000 timestamps per product per day.
- The notebook filters out `HYDROGEL_PACK` before computing product frequencies, so its printed denominator is 330,000 rather than the full 360,000.
- The target product trade counts match across the notebook and the longer EDA: `VELVETFRUIT_EXTRACT` has 1,372 trades; the most active vouchers are `VEV_4000`, `VEV_6000`, `VEV_6500`, `VEV_5500`, and `VEV_5400`.
- `VELVETFRUIT_EXTRACT` has a very stable top-of-book spread around 5 ticks, with bid and ask offsets centered around -2.5 and +2.5 from mid.
- Underlying one-tick log returns are non-normal and negatively autocorrelated at lag 1, but ten-tick returns look much closer to normal.
- The underlying has rolling annualized realized volatility around 0.41, while the ATM IV proxy sits around 0.235, producing a persistent IV-minus-RV gap near -0.178.
- The strongest gamma-scalp proxy appears in the near-ATM complex: `VEV_5200`, `VEV_5300`, `VEV_5400`, and `VEV_5100`.
- Deep OTM vouchers `VEV_6000` and `VEV_6500` are pinned at bid 0 / ask 1 / mid 0.5; they create large percentage spreads and fragile implied-volatility signals.
- Lead-lag correlations peak at lag 0 for all active vouchers. This argues against simple delayed cross-product arbitrage and toward delta/gamma fair-value modeling.
- No-arbitrage flags are concentrated in lower-bound violations for deep ITM `VEV_4000` and `VEV_4500`; these are likely driven by integer prices, near-zero extrinsic value, and midpoint conventions, not automatically executable arbitrage.

1 Dataset Structure From EDA.ipynb

Raw Shape And Product Coverage

- The full combined prices file is `phase2/round3/data/prices_round_3_combined.csv`.
- The full combined trades file is `phase2/round3/data/trades_round_3_combined.csv`.
- The price schema is: `day`, `timestamp`, `product`, three bid prices, three bid volumes, three ask prices, three ask volumes, `mid_price`, and `profit_and_loss`.
- The trade schema contains `timestamp`, `buyer`, `seller`, `symbol`, `currency`, `price`, `quantity`, and `day`.
- Every product appears at every 100-timestamp grid point from 0 to 999900 on each historical day.

Full combined prices:

HYDROGEL_PACK	30000
VELVETFRUIT_EXTRACT	30000
VEV_4000	30000
VEV_4500	30000
VEV_5000	30000
VEV_5100	30000
VEV_5200	30000
VEV_5300	30000
VEV_5400	30000
VEV_5500	30000
VEV_6000	30000
VEV_6500	30000

Full combined trades:

HYDROGEL_PACK	1010
VELVETFRUIT_EXTRACT	1372
VEV_4000	464
VEV_4500	1
VEV_5000	1
VEV_5100	1
VEV_5200	18
VEV_5300	121
VEV_5400	225
VEV_5500	267
VEV_6000	284
VEV_6500	284

Notebook Product-Frequency Print

- The notebook prints 30,000 rows per target product over 330,000 filtered rows.
- This is internally consistent after filtering, but it is not the full dataset denominator because `HYDROGEL_PACK` is absent from the filtered target list.

```
1. PRODUCT APPEARANCE FREQUENCIES IN ORDER BOOK
VEV_4000      : 30000/330000 (09.09%)
VEV_4500      : 30000/330000 (09.09%)
VEV_5000      : 30000/330000 (09.09%)
VEV_5100      : 30000/330000 (09.09%)
VEV_5200      : 30000/330000 (09.09%)
VEV_5300      : 30000/330000 (09.09%)
VEV_5400      : 30000/330000 (09.09%)
VEV_5500      : 30000/330000 (09.09%)
VEV_6000      : 30000/330000 (09.09%)
VEV_6500      : 30000/330000 (09.09%)
VELVETFRUIT_EXTRACT : 30000/330000 (09.09%)
```

Order Book Prices Distribution by Layer (L1 to L3)

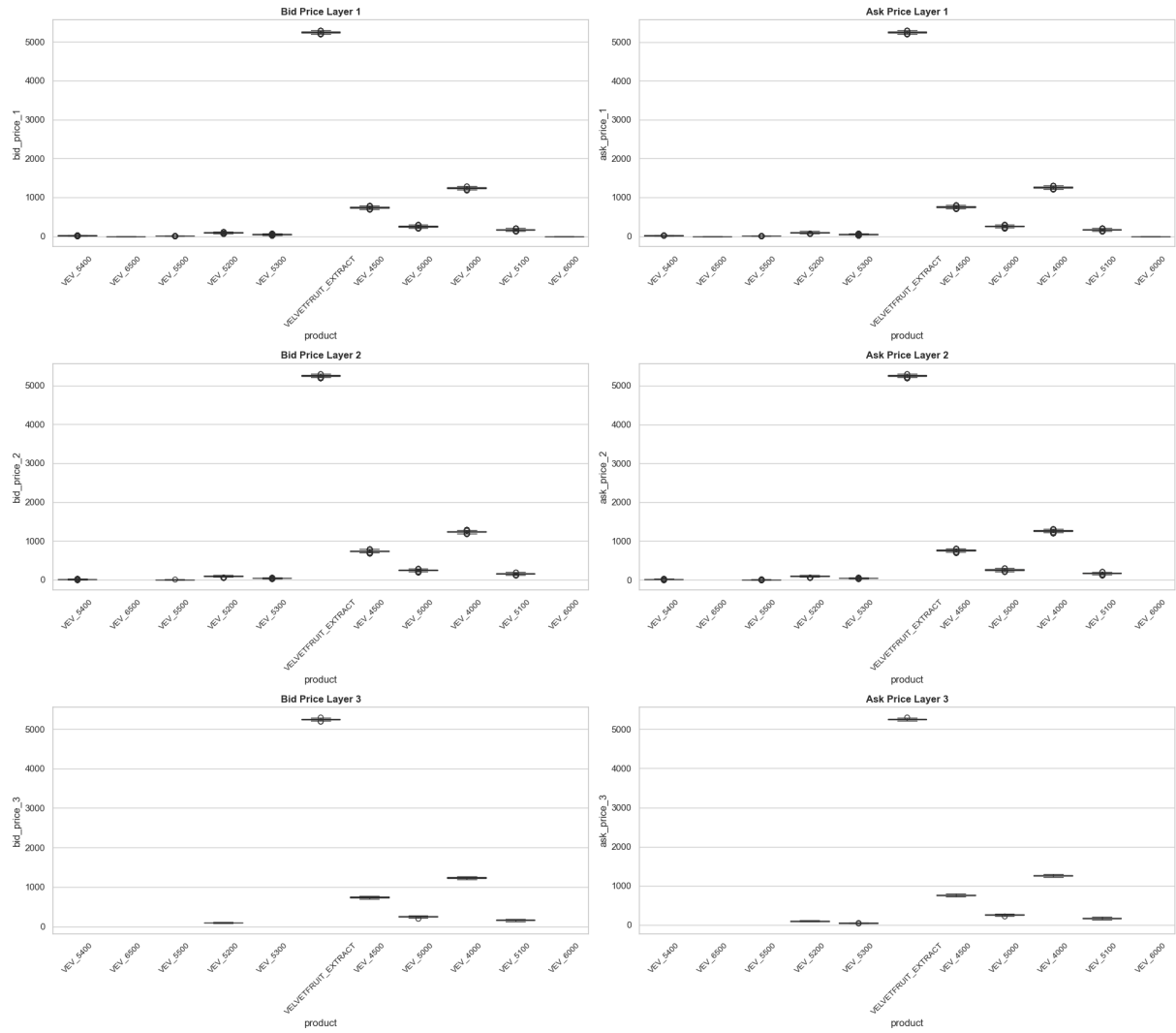


Figure 1: Notebook plot: order-book price distributions by product and layer.

Order Book Volumes Distribution by Layer (L1 to L3)

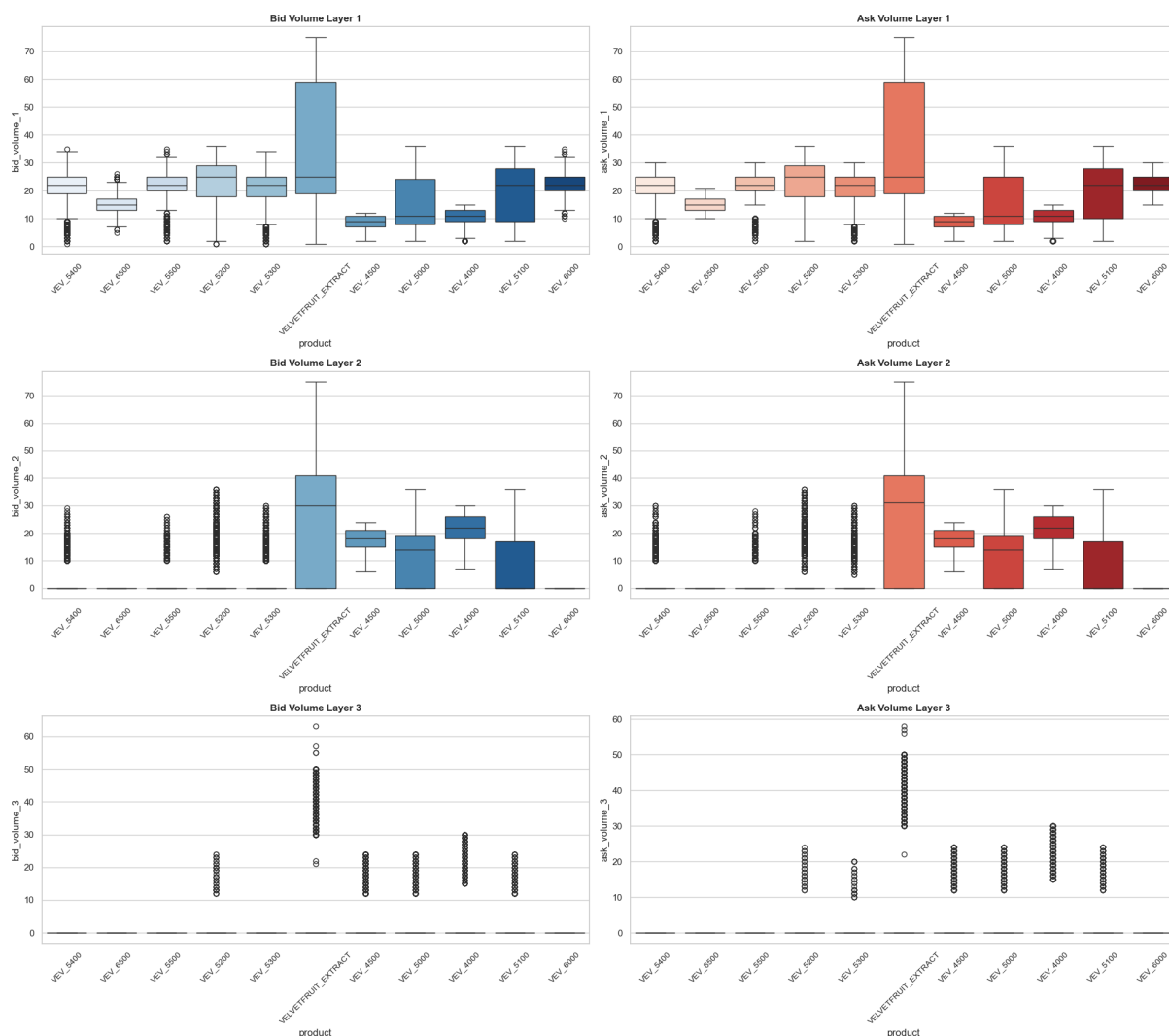


Figure 2: Notebook plot: order-book volume distributions by product and layer.

Order Book Structure Discoveries

- VELVETFRUIT_EXTRACT mid is centered near 5250 with top-of-book bid and ask means near 5247.6 and 5252.6.
- VEV_4000 and VEV_4500 behave like deep ITM calls: high absolute prices, large spreads, tiny extrinsic value.
- VEV_5200 and VEV_5300 are the most natural ATM candidates because spot is near 5250.
- VEV_6000 and VEV_6500 are pinned at bid 0, ask 1, mid 0.5 for all timestamps.
- Top-level voucher volumes are stable and integer quoted; the more OTM vouchers have increasingly shallow deeper-book availability.
- Average tick distance between order-book appearances is exactly 1 for all target products, meaning every filtered product has a quote at every timestamp grid point.

Mean L1 view from notebook:

Product	bid_price_1	bid_volume_1	ask_price_1	ask_volume_1
VEV_4000	1239.70	10.92	1260.52	10.92
VEV_4500	742.18	8.95	758.04	8.95
VEV_5000	252.00	15.33	258.04	15.48

VEV_5100	164.66	19.25	168.95	19.35
VEV_5200	94.10	22.56	96.99	22.51
VEV_5300	45.71	20.28	47.81	20.24
VEV_5400	15.26	21.76	16.64	21.72
VEV_5500	6.07	22.18	7.22	22.17
VEV_6000	0.00	22.48	1.00	22.48
VEV_6500	0.00	15.48	1.00	15.48
VELVETFRUIT_EXTRACT	5247.60	37.83	5252.59	37.80

Advanced Order Book Metrics Distributions

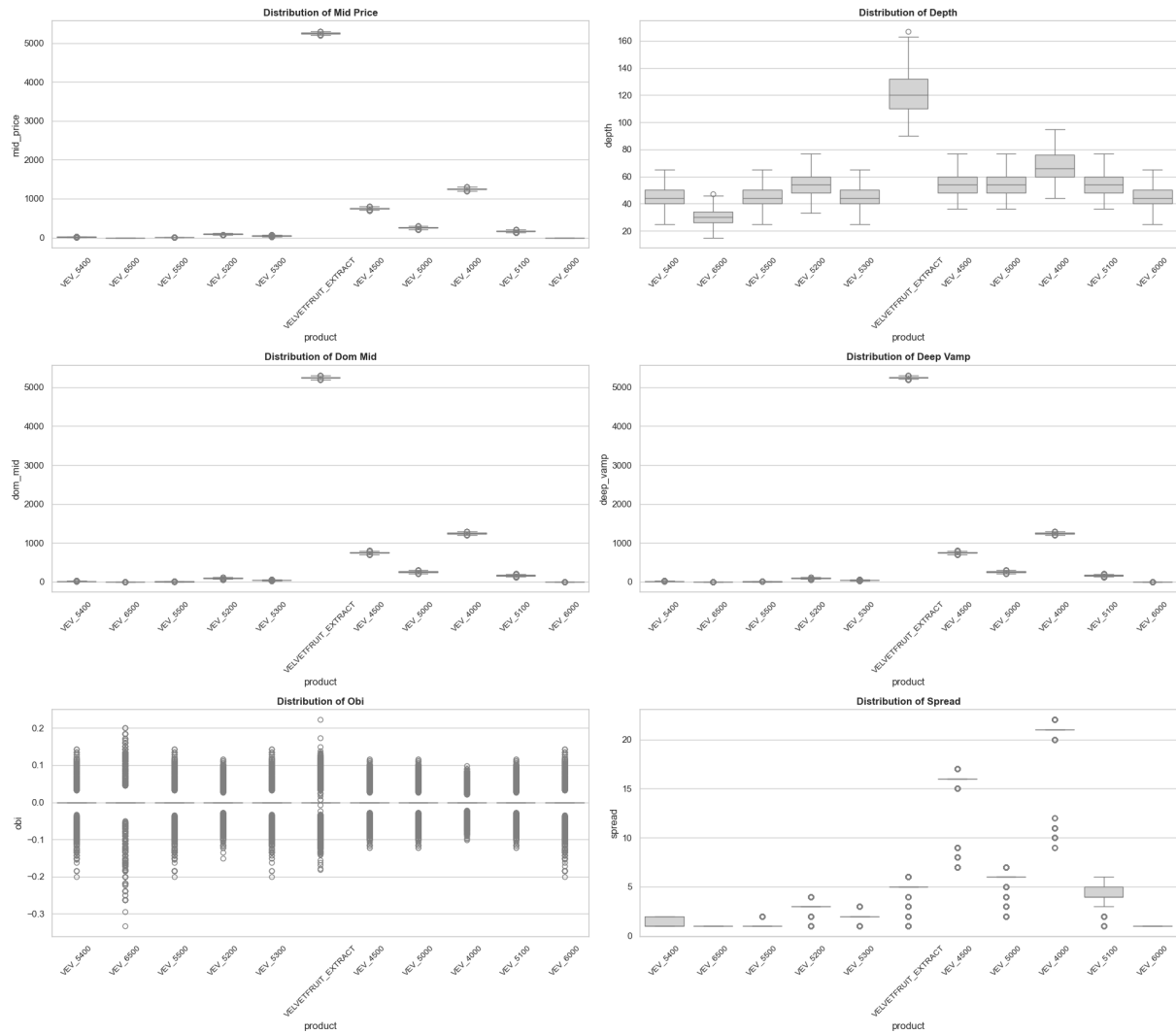


Figure 3: Notebook plot: advanced order-book metric distributions.

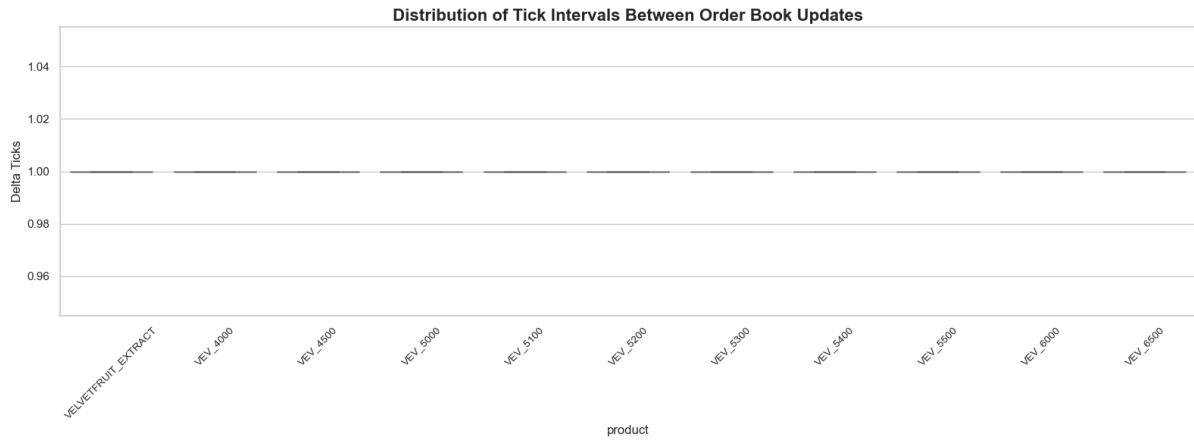


Figure 4: Notebook plot: tick distance between order-book appearances.

Notebook Trade And Depletion Findings

- VELVETFRUIT_EXTRACT trades far more frequently than any voucher.
- Among vouchers, VEV_4000 trades often but with tiny size around 2; near-ATM and OTM active vouchers trade in sizes around 3.5.
- VEV_4500, VEV_5000, and VEV_5100 only have one trade each, all on day 1.
- VEV_5200 is sparse, with 18 total trades across three days.
- The notebook's depletion scan finds 188 underlying events consuming at least layer 1, and only one event consuming at least layer 2/3: VEV_6500 on day 0 at timestamp 640300.
- No empty L1, one-sided L1, posted-L2-without-L1, or posted-L3-without-L2 anomalies are found after the notebook fills missing volumes with zero.

Trades Count per Product per Day:

day	VEV_4000	VEV_4500	VEV_5000	VEV_5100	VEV_5200	VEV_5300	VEV_5400	VEV_5500	VEV_6000	VEV_6500	VELVETFRUIT_EXTRACT
0	172	0	0	0	3	37	64	81	91	91	445
1	164	1	1	1	7	39	81	92	98	98	450
2	128	0	0	0	8	45	80	94	95	95	477

LOB depletion:

Instances consuming AT LEAST Layer 1:

VEV_5200	7
VEV_5300	19
VEV_5400	5
VEV_5500	5
VEV_6500	1
VELVETFRUIT_EXTRACT	188

Instances consuming AT LEAST Layer 2 and Layer 3:

day=0, timestamp=640300, product=VEV_6500, total_trade_qty=5, avg_trade_price=0.0

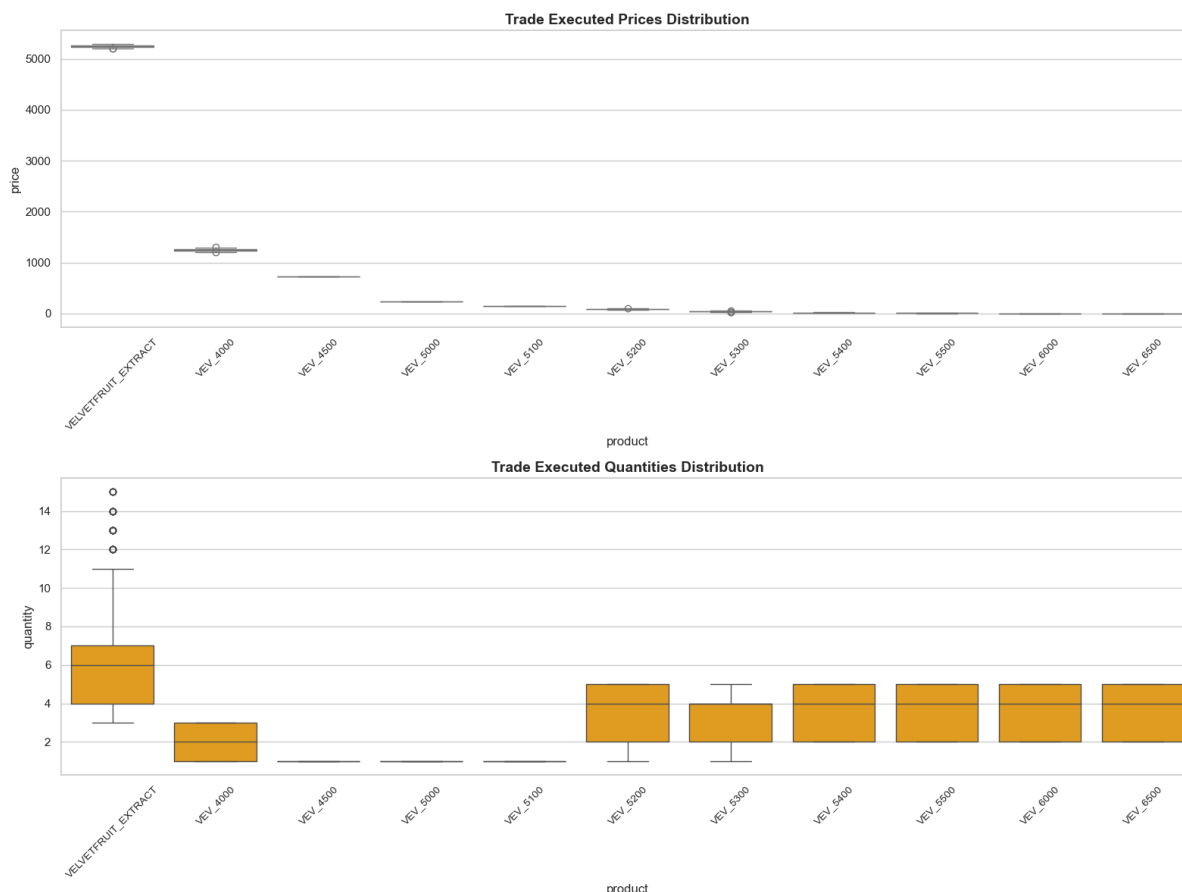


Figure 5: Notebook plot: trade price and trade quantity distributions.

2 Data Integrity From The Longer EDA

- The longer EDA uses per-day files in `phase2/round3/algo/data`; these match the combined target-product counts from the notebook.
- Every target product has 10,000 book snapshots per day and no timestamp gaps.
- No crossed books, missing mids, duplicate timestamps, or trades without same-timestamp book rows are reported.
- Six VELVETFRUIT_EXTRACT trades are outside the visible top bid/ask across all days; this should be interpreted as hidden depth or timing artifact before being treated as edge.
- Deeper book availability varies sharply: VEV_4000 and VEV_4500 almost always have deeper levels, while VEV_6000 and VEV_6500 are L1-only.

Data integrity highlights:

VELVETFRUIT_EXTRACT: 10000 book snapshots per day, trades 445/450/477, outside top book 1/3/2.
 VEV_4000: 10000 book snapshots per day, trades 172/164/128, no top-book anomalies.
 VEV_5200: 10000 book snapshots per day, trades 3/7/8, no top-book anomalies.
 VEV_6000: 10000 book snapshots per day, trades 91/98/95, L1-only book.
 VEV_6500: 10000 book snapshots per day, trades 91/98/95, L1-only book.

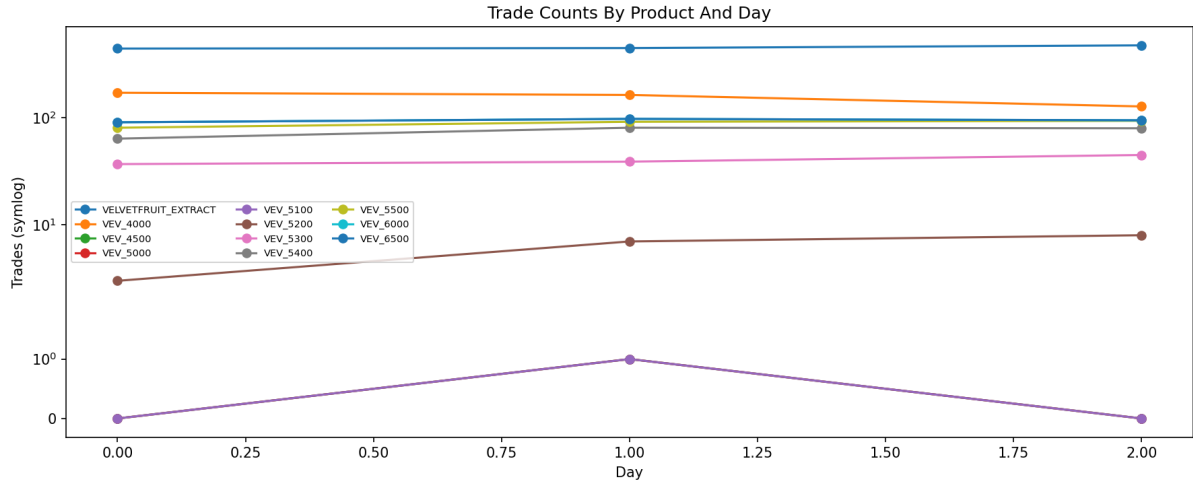


Figure 6: Long EDA: trade counts by product and day.

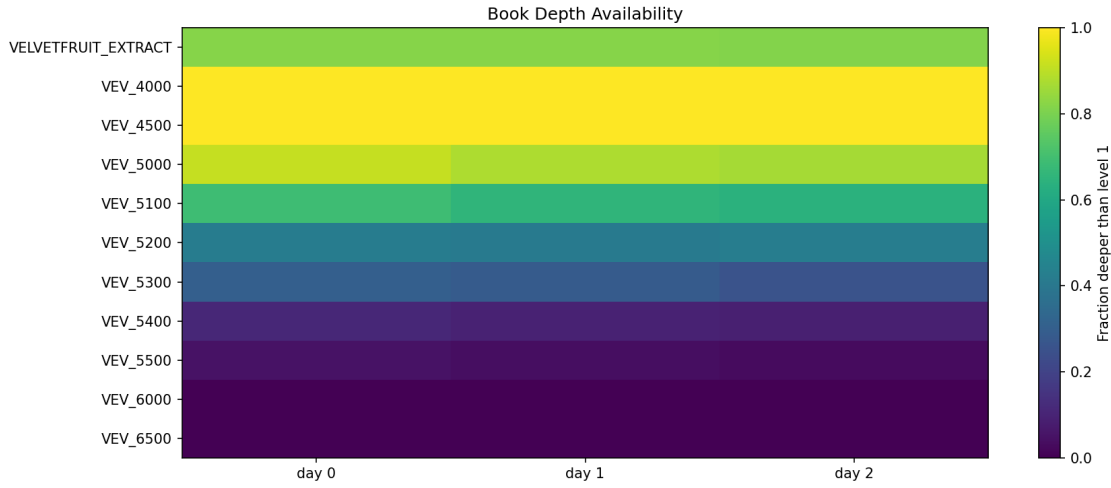


Figure 7: Long EDA: deeper-book availability by product and day.

3 Underlying: VELVETFRUIT_EXTRACT

Price, Return, And Volatility Findings

- Mid price oscillates around 5250 with visible intraday trends and day-to-day continuation.
- Tick-level price-change standard deviation is about 1.12 to 1.14 price units.
- Day-level drift is small but positive on days 1 and 2: about +20.5 and +28.0 ticks per day, versus about -6.0 on day 0.
- Around 24.5% of tick-level mid changes are exactly zero; positive and negative changes are otherwise balanced.
- One-tick log returns reject normality very strongly; ten-tick returns have much weaker non-normality and much higher Jarque-Bera p-values.
- Rolling realized volatility is stable: RV500 is near 0.410 to 0.414 by day.
- ACF of mid price is near 0.997 at lag 1, reflecting price-level persistence; ACF of one-tick returns is -0.159 at lag 1, reflecting bid/ask bounce or short-horizon mean reversion.

Return stats:

day	horizon	mean	std	skew	kurtosis	JB p-value
0	1	-1.14e-07	0.000214	-0.0397	0.4517	9.26e-20
1	1	3.90e-07	0.000216	-0.0307	0.3246	1.33e-10
2	1	5.30e-07	0.000217	-0.0294	0.2828	2.82e-08
pooled 10		3.08e-06	0.000576	-0.0024	-0.0073	0.9535

Stationarity diagnostics:
ADF p-value = 4.71e-05
KPSS p-value = 0.01
Variance ratios: VR(2)=0.841, VR(5)=0.748, VR(10)=0.715, VR(20)=0.705
Hurst R/S = 0.999
Script classification = trending

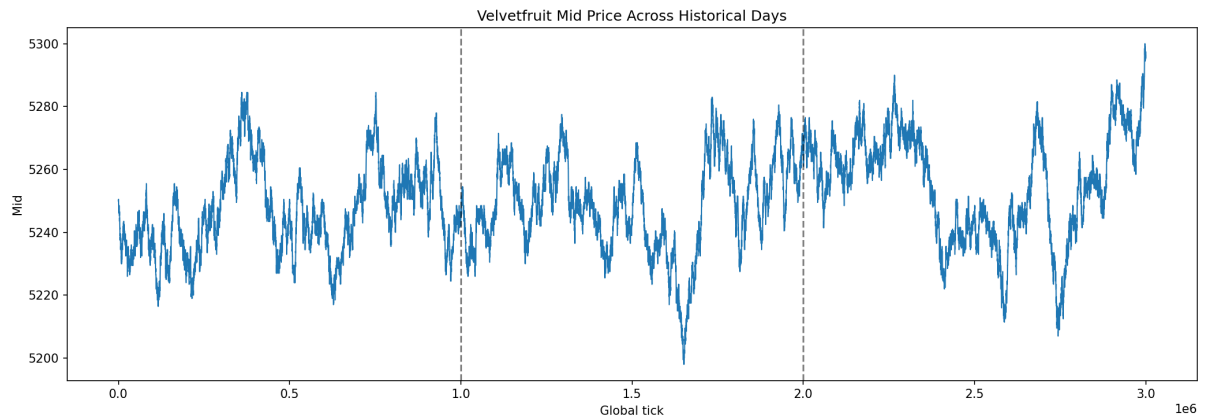


Figure 8: Underlying mid price across historical days.

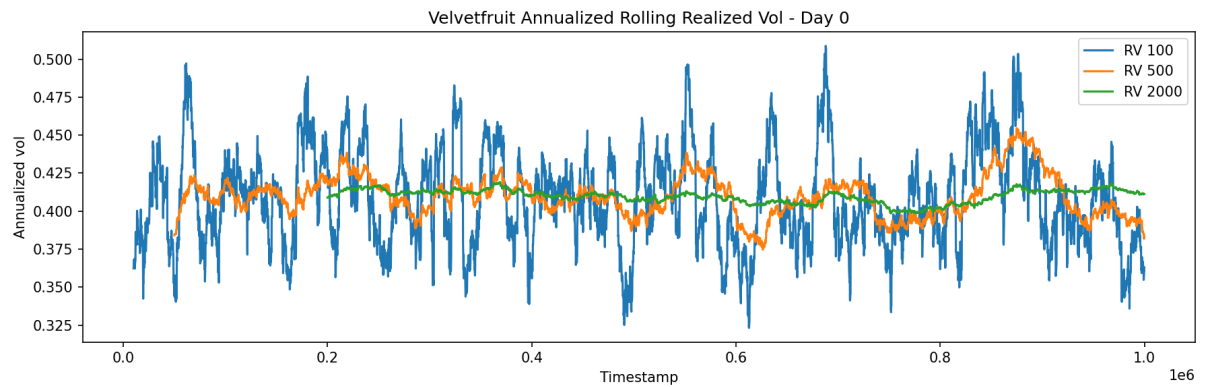


Figure 9: Underlying rolling realized volatility on day 0.

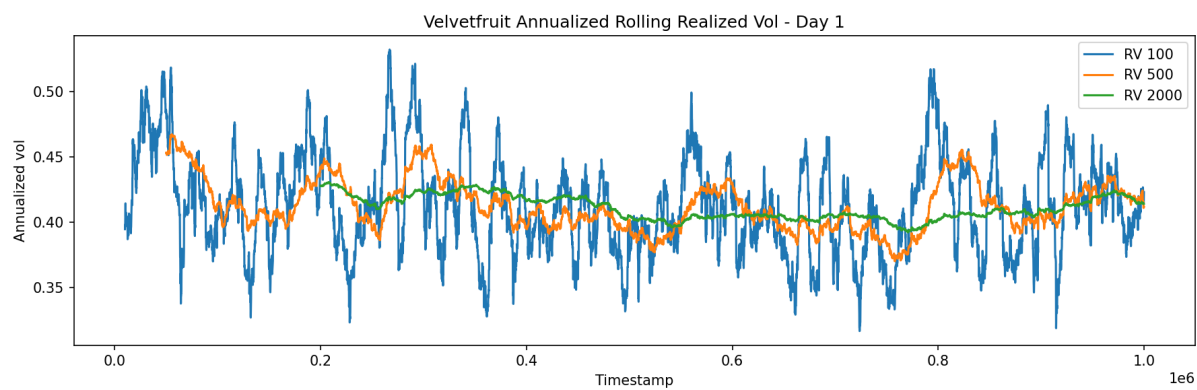


Figure 10: Underlying rolling realized volatility on day 1.

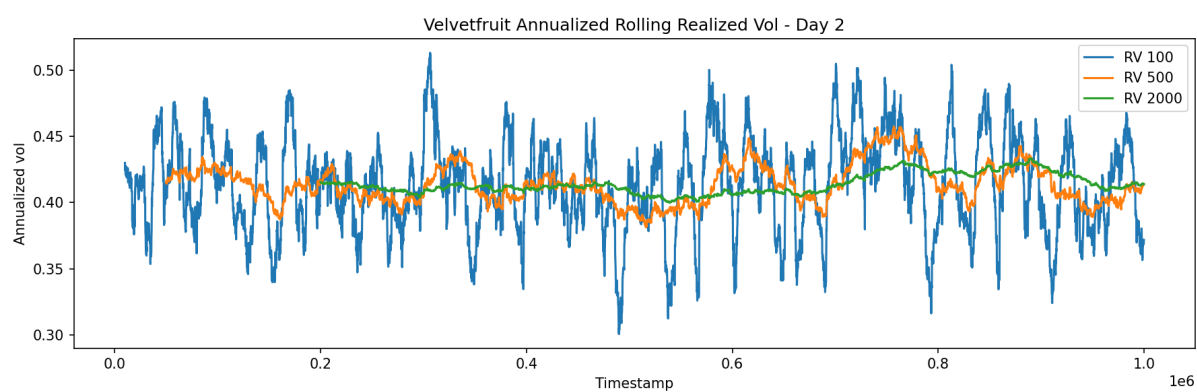


Figure 11: Underlying rolling realized volatility on day 2.

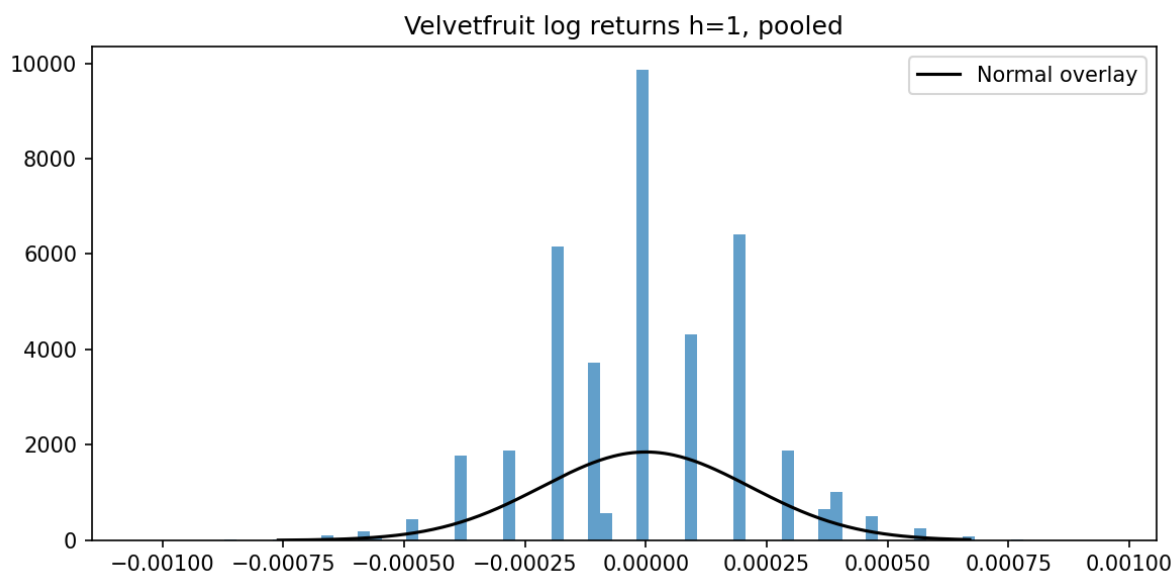


Figure 12: Pooled one-tick log returns.

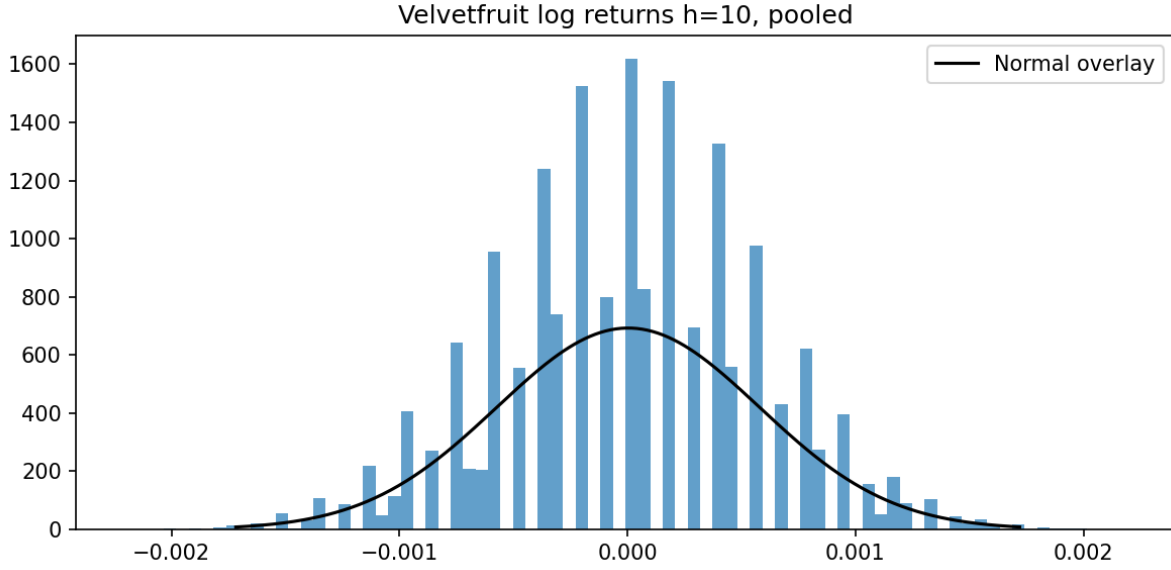


Figure 13: Pooled ten-tick log returns.

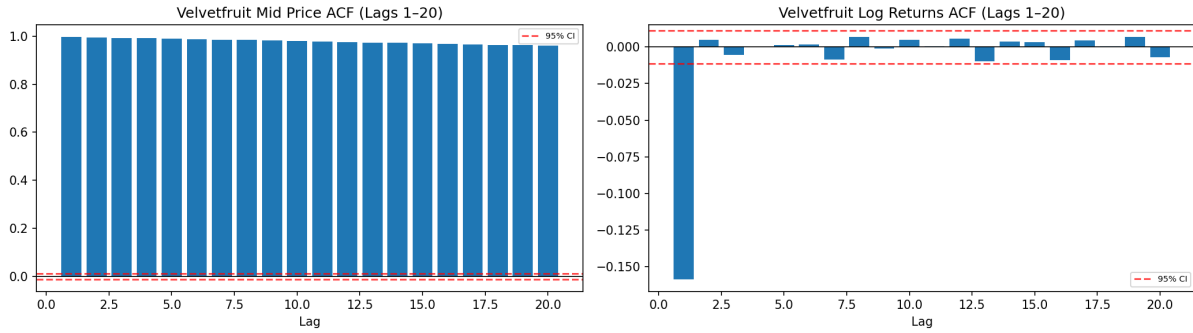


Figure 14: Underlying price and return autocorrelation.

Liquidity, Flow, And Execution Findings

- The underlying spread is highly stable: mean 4.988, median 5, p95 6.
- Bid and ask quote offsets are symmetric around mid: -2.494 and +2.494 on average.
- Top L1 depth averages about 37.8 units per side, and L2/L3 average around 40 per side when present.
- Trade arrival is close to Poisson-like by 1000-timestamp buckets; dispersion indices are between 0.86 and 1.00.
- Trades are classified as 781 buys and 591 sells using trade price versus mid.
- Buy trades are followed by positive forward log returns at horizons 100, 500, and 1000; sell trades are weaker and only clearly negative at 100 and 1000.
- Maker fills are buy-flow biased: trades at ask are 55.3% to 58.7% of underlying prints by day.
- Crossing the spread costs about 2.494 ticks versus mid per taker trade, so naive taker hedging needs substantial option edge or urgent risk control.

Spread summary:

day	mean	median	mode	p95
0	4.9935	5.0	5	6.0
1	4.9846	5.0	5	6.0
2	4.9863	5.0	5	6.0

Flow toxicity:

horizon	side	mean_forward_return	count
100	buy	0.000079	781
100	sell	-0.000023	591
500	buy	0.000089	781
500	sell	0.000002	591
1000	buy	0.000075	781
1000	sell	-0.000030	591

Maker fills:

day	ask-side fills	bid-side fills	ask-fill fraction
0	252	193	0.566
1	249	201	0.553
2	280	197	0.587

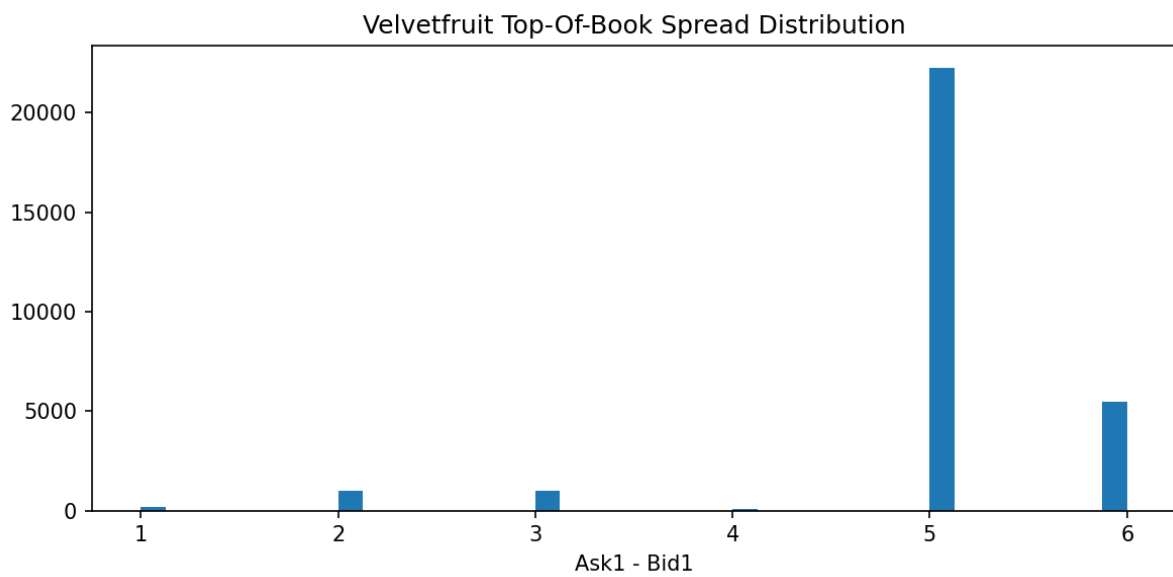


Figure 15: Underlying top-of-book spread distribution.

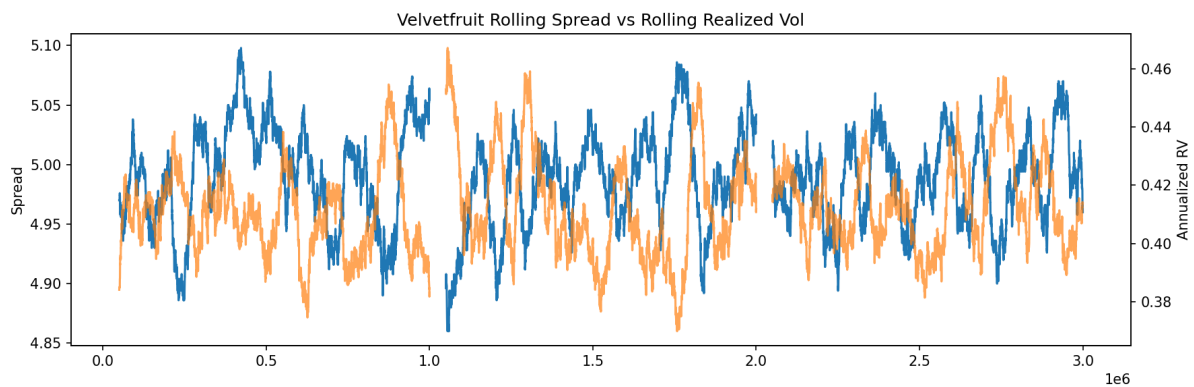


Figure 16: Rolling spread versus rolling realized volatility.

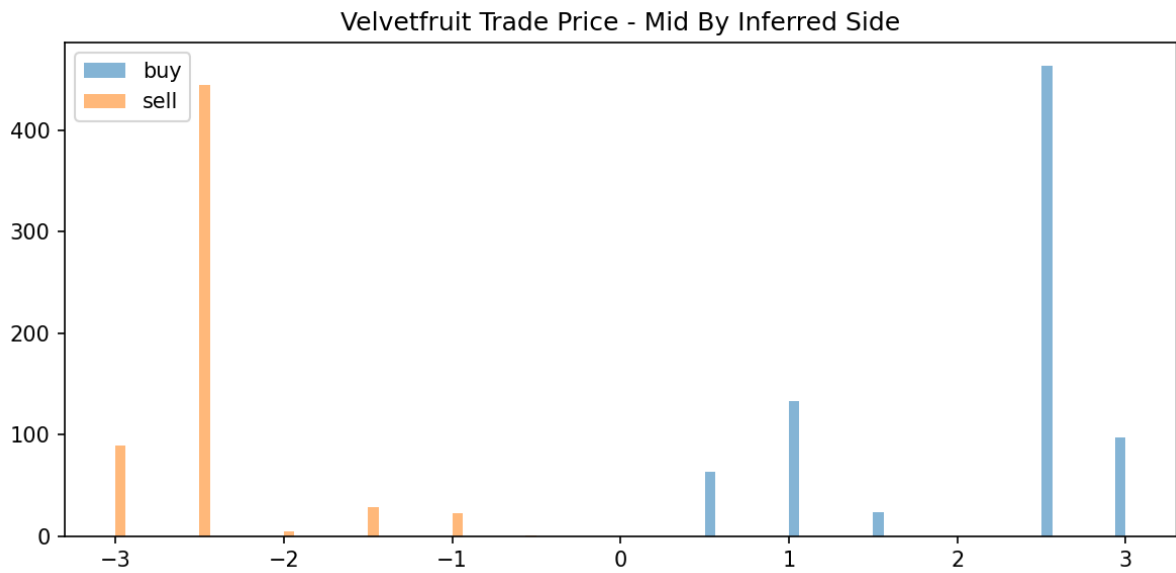


Figure 17: Underlying trade price minus mid by inferred side.

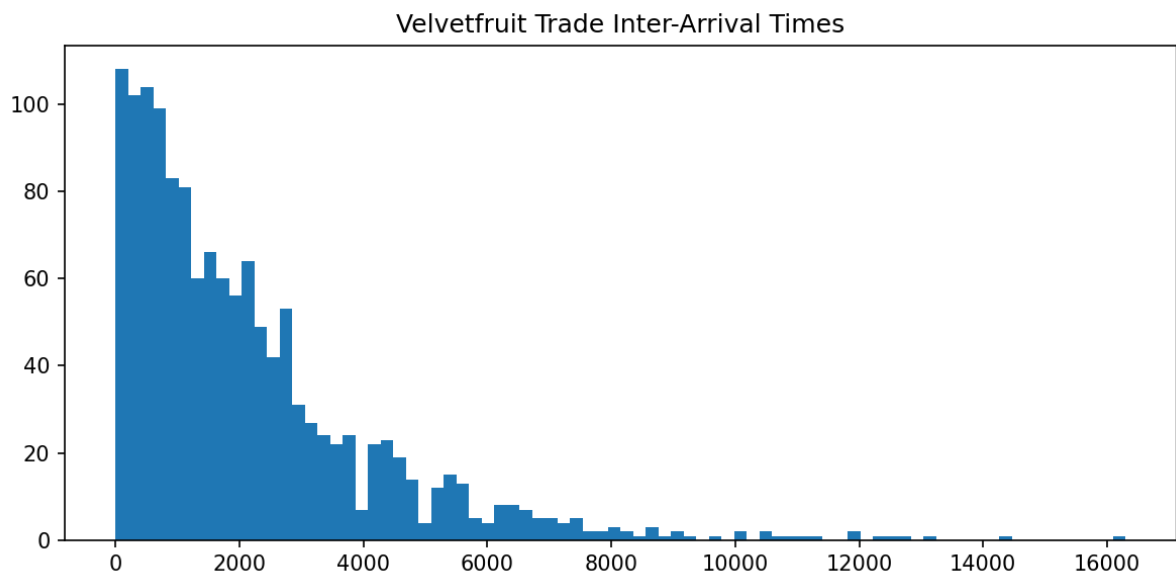


Figure 18: Underlying trade interarrival distribution.

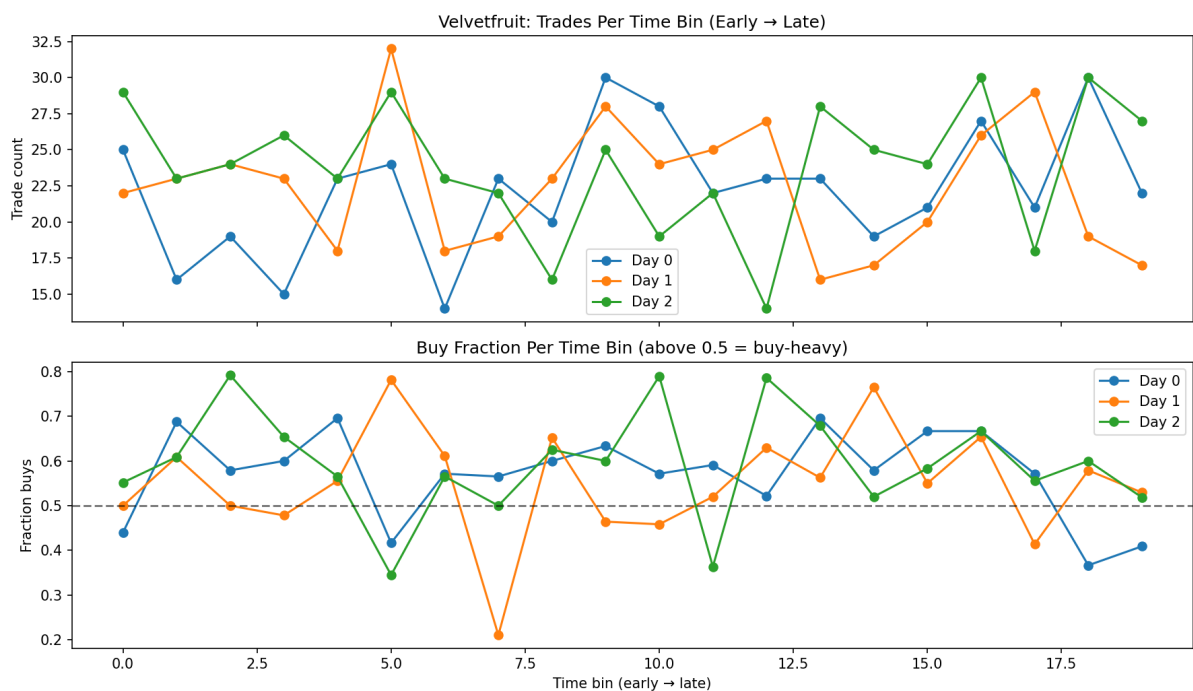


Figure 19: Underlying intraday trade count and buy fraction.

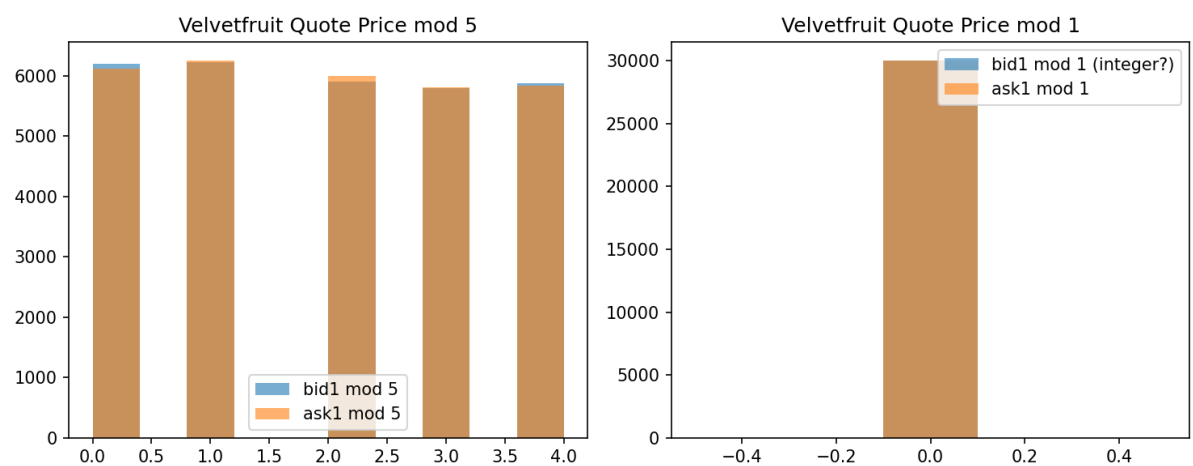


Figure 20: Underlying quote rounding.

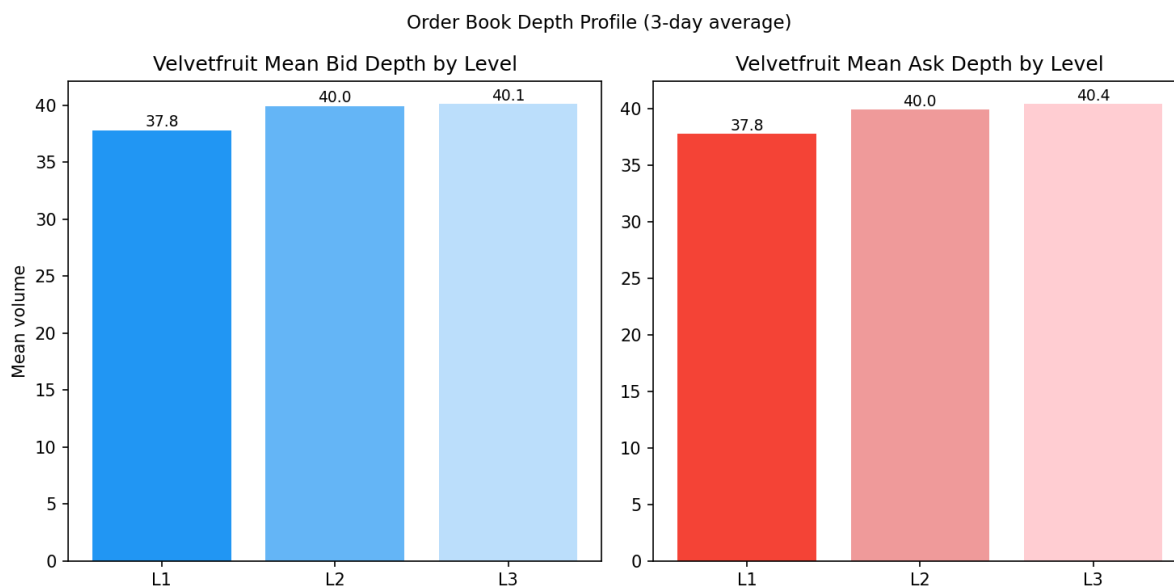


Figure 21: Underlying depth profile.

4 Voucher Microstructure And Moneyness

- VEV_4000, VEV_4500, and VEV_5000 are ITM across all days, although VEV_5000 is closer to the money.
- VEV_5100 transitions from mostly near-ATM to more ITM by day 2.
- VEV_5200 and VEV_5300 are the cleanest near-ATM strikes.
- VEV_5400 is near-ATM but partly OTM, especially early.
- VEV_5500, VEV_6000, and VEV_6500 are OTM across all days.
- Absolute spreads decline as strike rises, but spread as a percentage of mid explodes in OTM floor-price vouchers.
- VEV_6000 and VEV_6500 have spread-to-mid equal to 2.0 because mid is 0.5 and spread is 1.
- VEV_4000 has many trades but wide absolute spreads around 21 ticks.
- VEV_5300, VEV_5400, and VEV_5500 are more relevant for active option market-making because they combine nonzero trade activity with lower absolute spreads.

Voucher liquidity by day:

VEV_4000: mean spread about 20.8, mean mid 1246-1255, trades/day 172/164/128.
 VEV_5200: mean spread about 2.86-2.93, mean mid 94-97, trades/day 3/7/8.
 VEV_5300: mean spread about 2.05-2.16, mean mid 44-49, trades/day 37/39/45.
 VEV_5400: mean spread about 1.33-1.43, mean mid 13.7-18.5, trades/day 64/81/80.
 VEV_5500: mean spread about 1.12-1.18, mean mid 5.3-8.1, trades/day 81/92/94.
 VEV_6000 and VEV_6500: mean spread 1, mean mid 0.5, trades/day near 91-98.

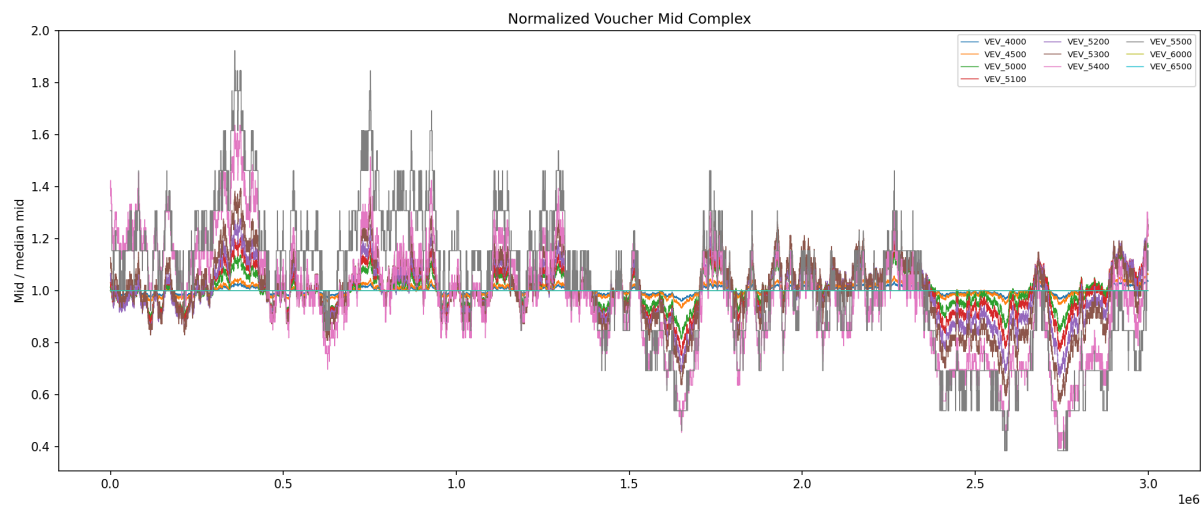


Figure 22: Normalized voucher mid-price complex.

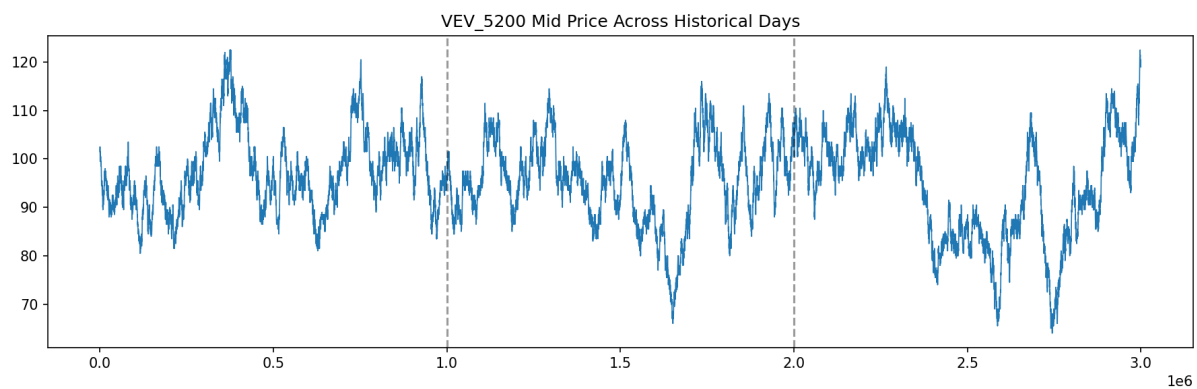


Figure 23: Representative near-ATM voucher mid: VEV 5200.

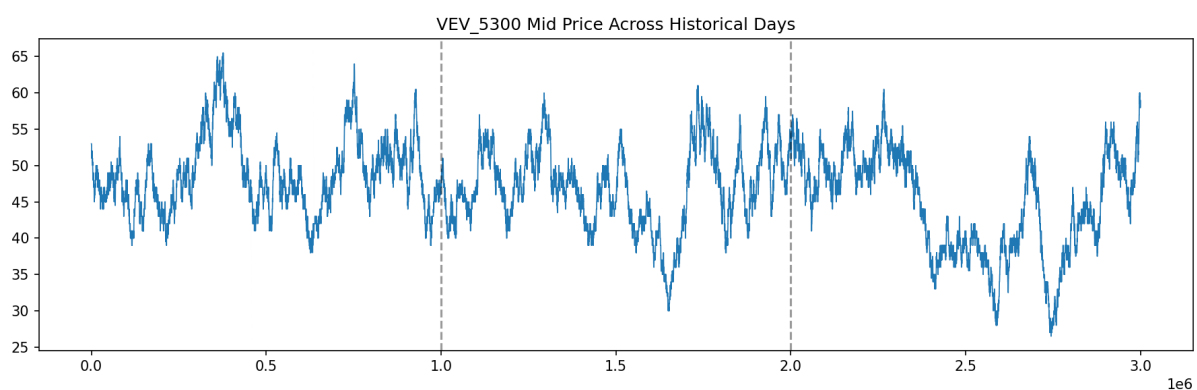


Figure 24: Representative near-ATM voucher mid: VEV 5300.

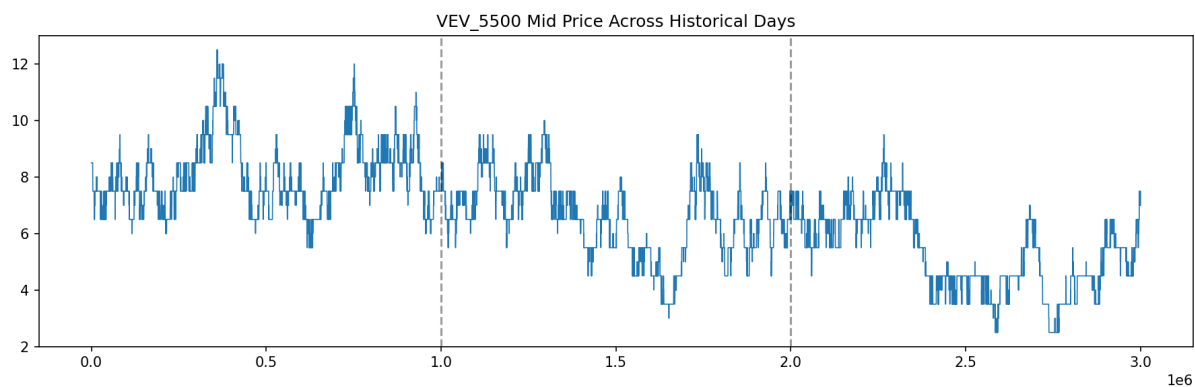


Figure 25: Representative OTM active voucher mid: VEV 5500.

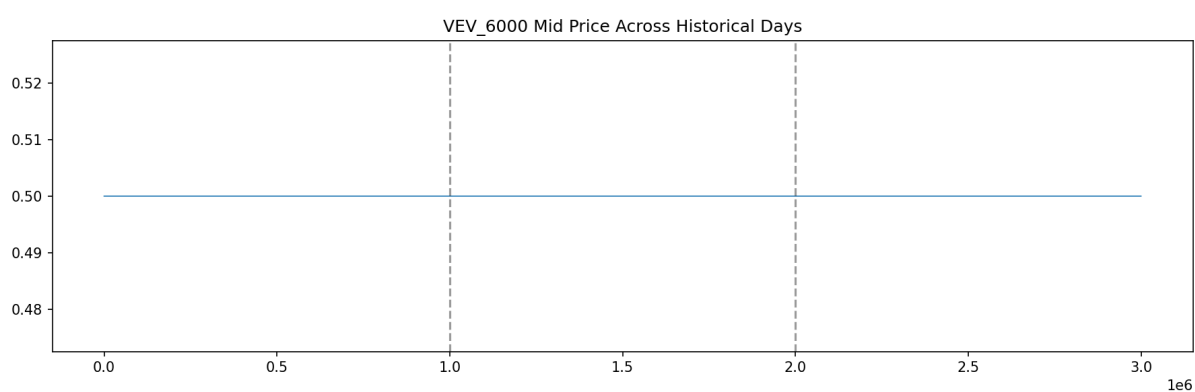


Figure 26: Pinned floor-price voucher mid: VEV 6000.

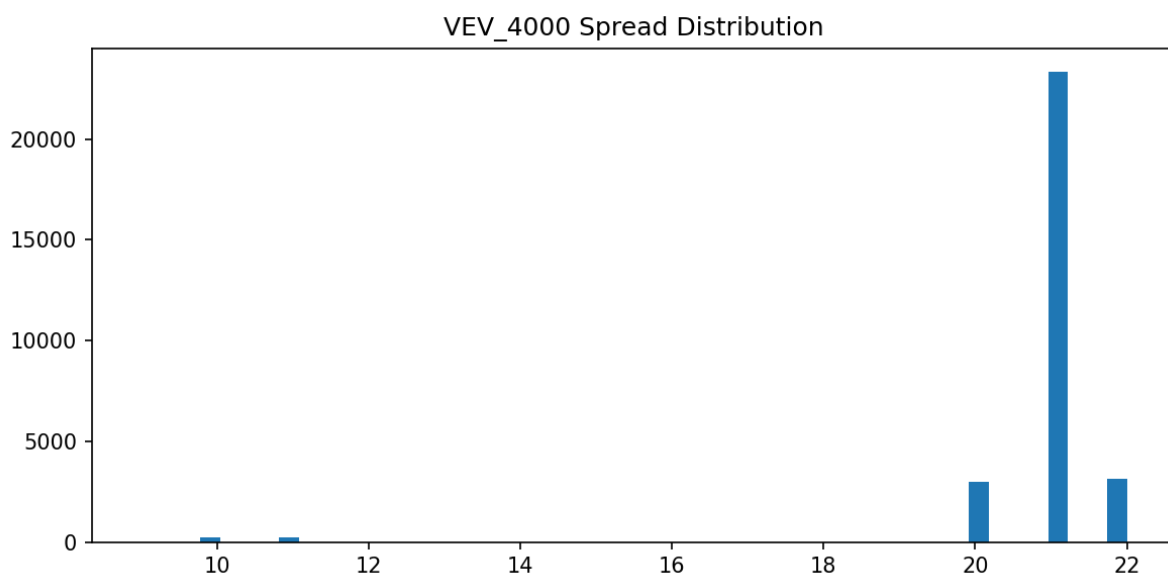


Figure 27: Deep ITM voucher spread distribution.

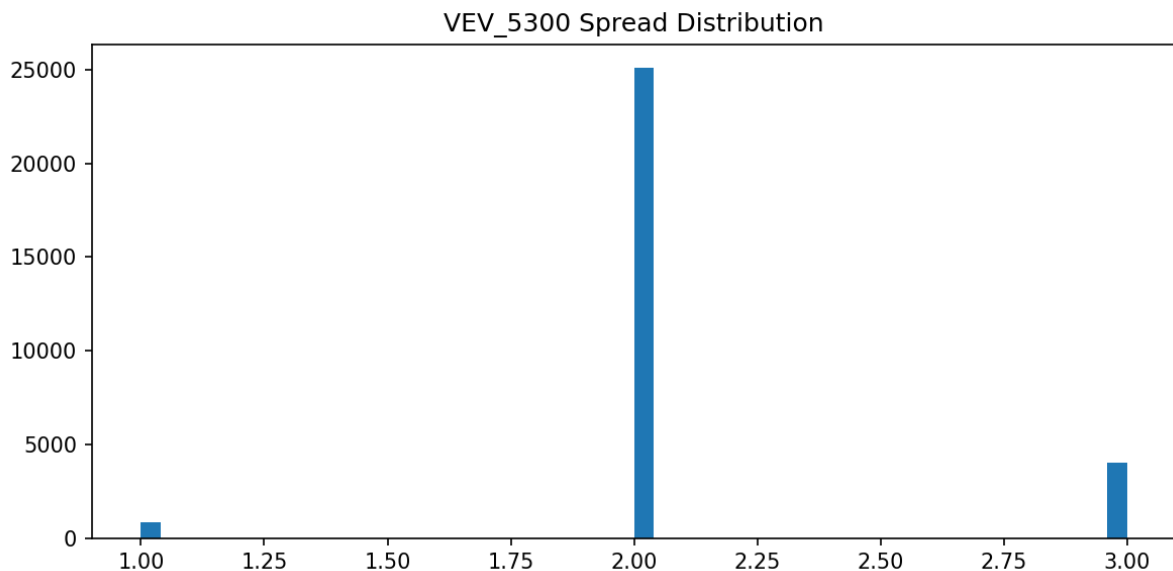


Figure 28: Near-ATM voucher spread distribution.

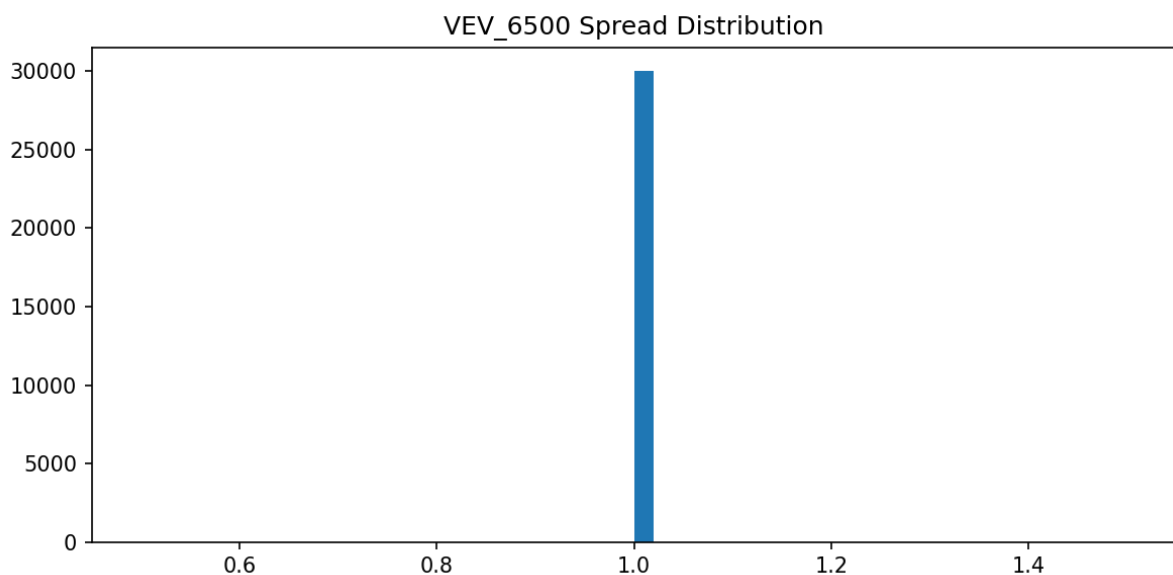


Figure 29: Deep OTM floor-price voucher spread distribution.

5 Options Valuation, IV, Greeks, And Arbitrage

Black-Scholes And IV Findings

- The longer EDA assumes Black-Scholes calls with zero risk-free rate and $T_years = TTE_days / 365$.
- Historical TTE mapping is day 0 = 8 days, day 1 = 7 days, day 2 = 6 days; live Round 3 is expected to have 5 days to expiry.
- ATM IV is selected as the closest valid-IV voucher at each timestamp; this is almost always VEV_5200 or VEV_5300.
- ATM IV has mean 0.232 on day 0, 0.237 on day 1, and 0.235 on day 2.

- RV500 has mean 0.410, 0.412, and 0.414 on days 0, 1, and 2.
- Therefore IV is persistently below realized volatility by about 17.5 to 18.0 vol points.
- The IV smile is smooth: quadratic smile fit mean R^2 is about 0.993, 0.989, and 0.987 across the three days.
- VEV_6000 and VEV_6500 have high implied vols but they are mostly artifacts of the 0/1 quote floor and should be treated carefully.

ATM IV versus RV500:

day	mean ATM IV	mean RV500	mean IV - RV
0	0.232175	0.409696	-0.177877
1	0.237291	0.412265	-0.175404
2	0.234733	0.413823	-0.179891

IV status counts:

ok	251785
zero_time_value	36414
below_lower_bound	11801

Main invalid-IV sources:

VEV_4000: below_lower_bound 2986, zero_time_value 24028.
VEV_4500: below_lower_bound 8812, zero_time_value 12365.
VEV_5000: below_lower_bound 3, zero_time_value 21.

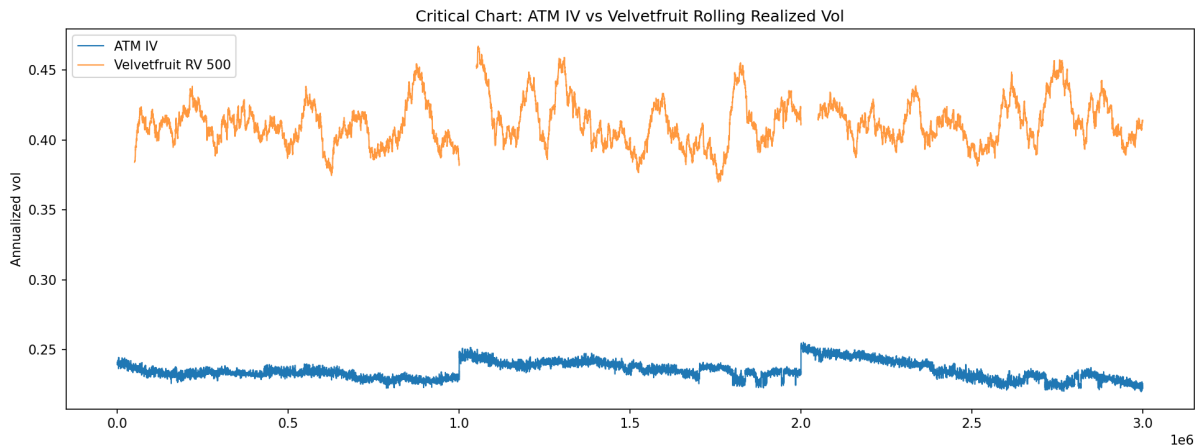


Figure 30: Critical chart: ATM IV versus underlying rolling realized volatility.

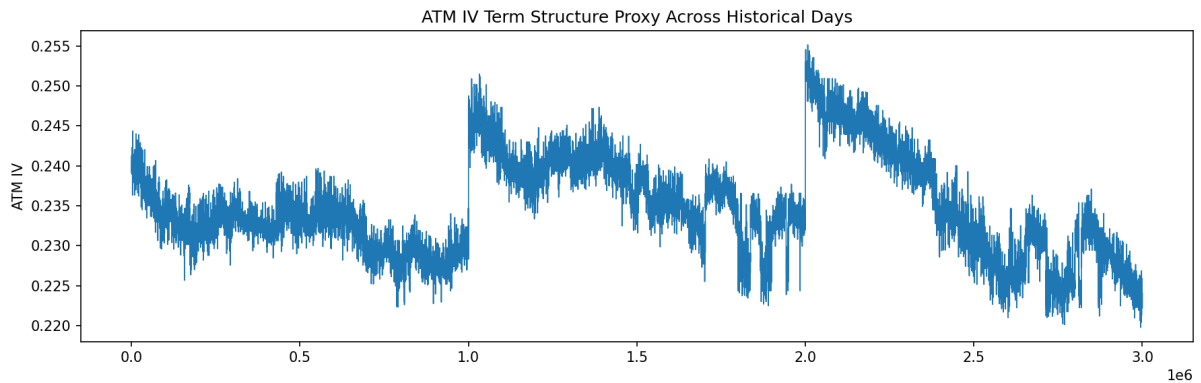


Figure 31: ATM IV proxy across historical days.

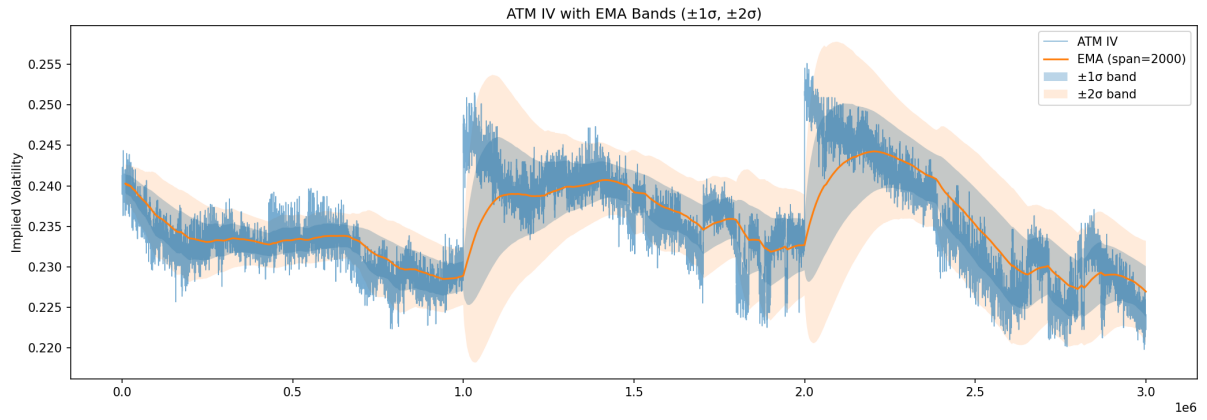


Figure 32: ATM IV with EMA bands.

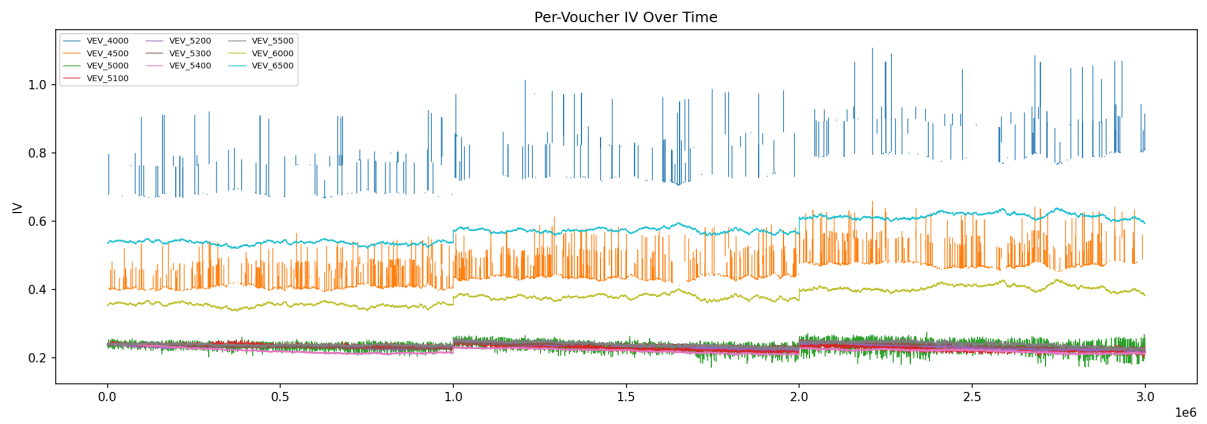


Figure 33: Per-voucher implied volatility over time.

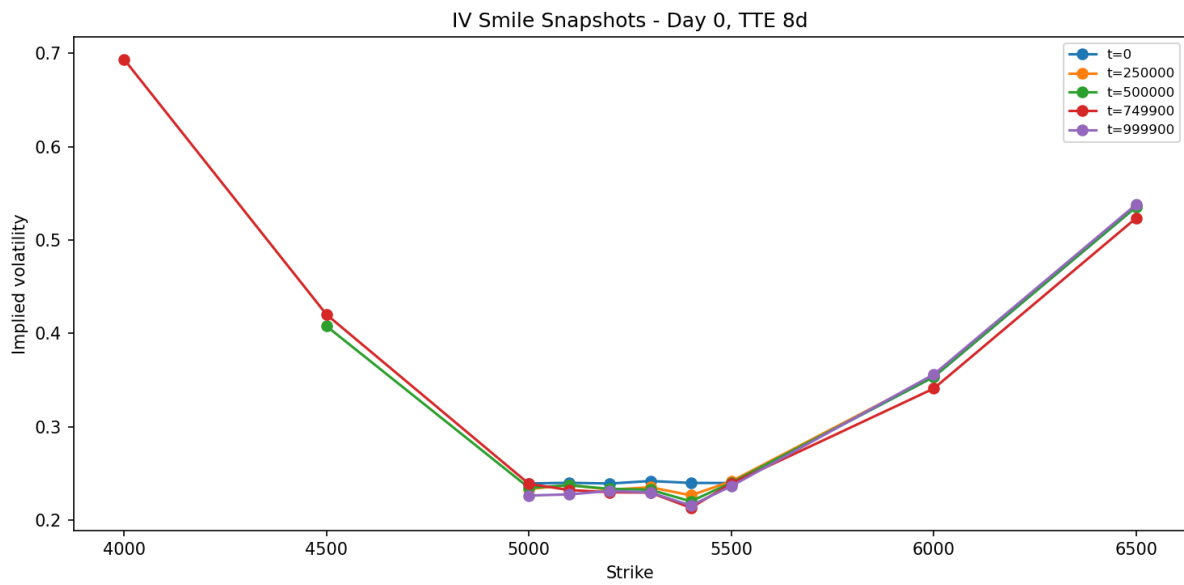


Figure 34: IV smile snapshots, day 0.

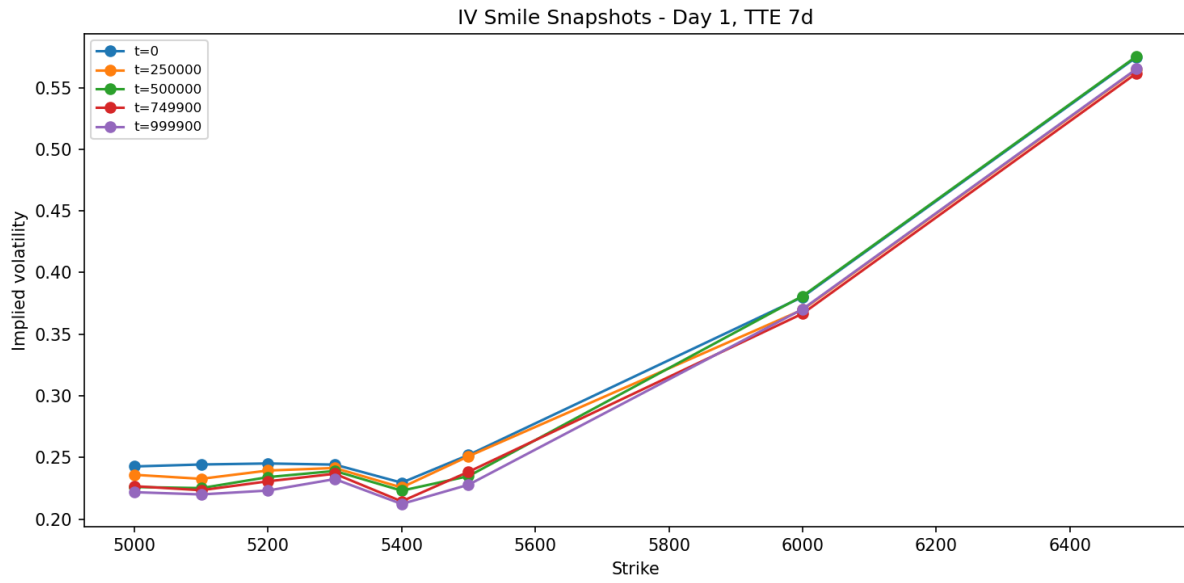


Figure 35: IV smile snapshots, day 1.

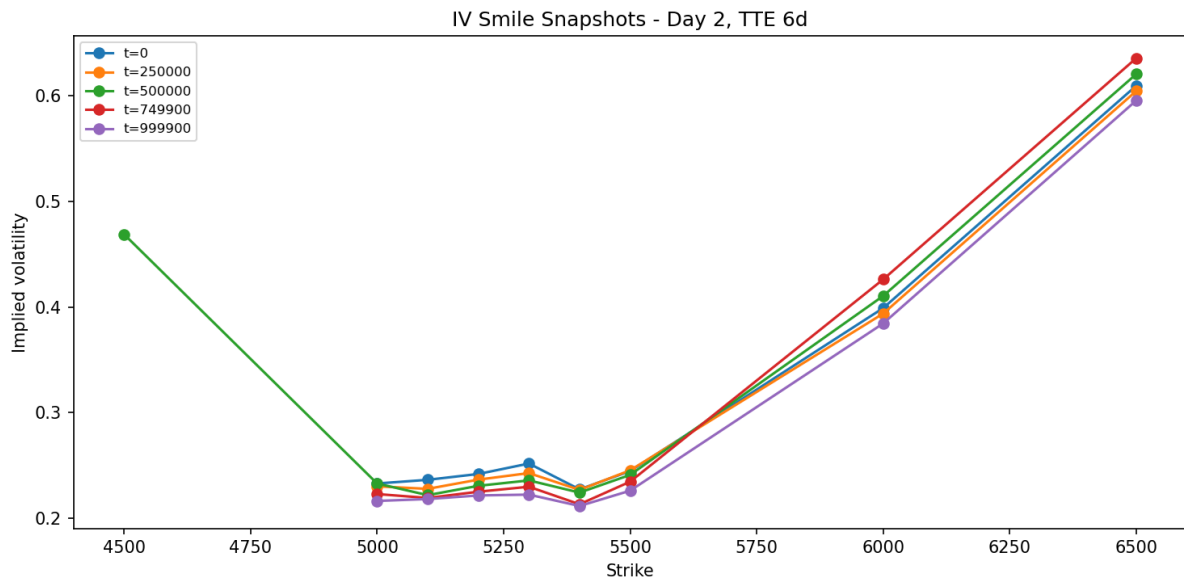


Figure 36: IV smile snapshots, day 2.

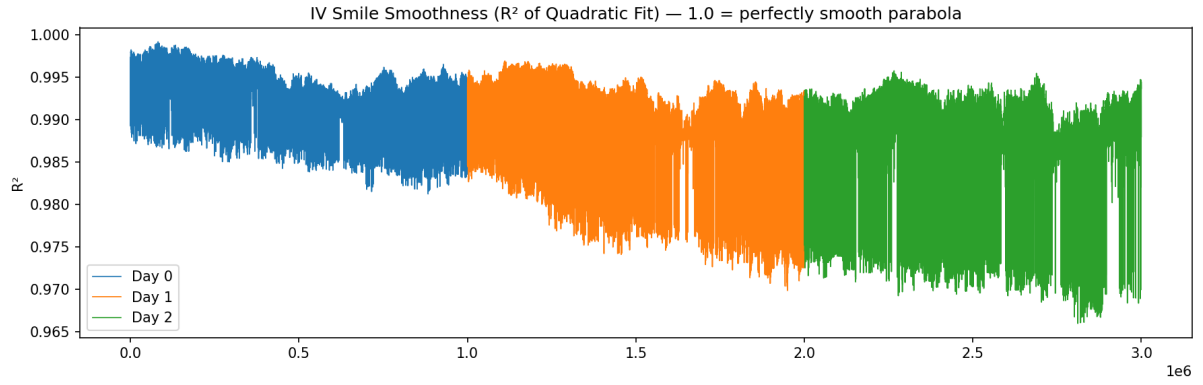


Figure 37: Quadratic IV-smile smoothness over time.

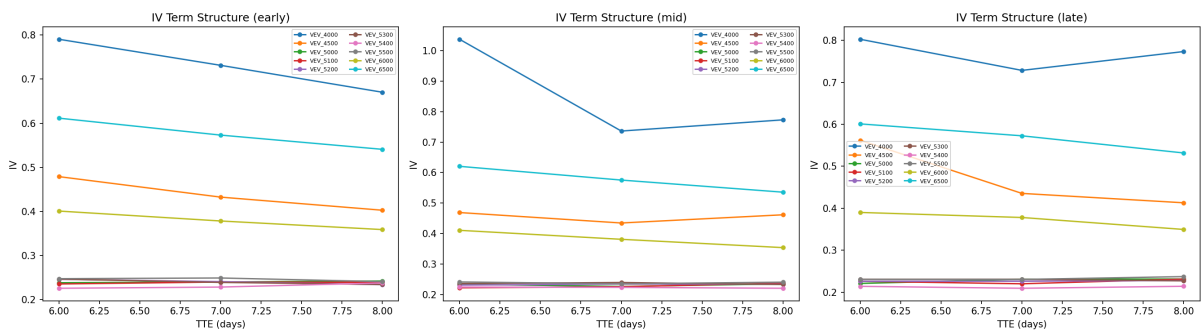


Figure 38: Cross-day IV term-structure proxy.

Greeks, Intrinsic Value, And Gamma-Scalp Proxy

- Median-sample delta declines monotonically from near 1.0 in deep ITM strikes to near zero in deep OTM strikes.
- Gamma is concentrated around VEV_5200 and VEV_5300, with VEV_5100 and VEV_5400 also relevant.
- VEV_4000 and VEV_4500 have almost no usable gamma despite high trade count or high notional price.
- Intrinsic/extrinsic decomposition confirms that VEV_4000 and VEV_4500 are almost pure intrinsic value.
- Extrinsic value decays with TTE in VEV_5000, VEV_5100, VEV_5200, and OTM strikes.
- Gamma-scalp proxy is positive for VEV_4500 through VEV_6000, with the largest proxy in VEV_5200 and VEV_5300.
- The proxy is negative for VEV_4000 and VEV_6500, where theta-like cost dominates realized gamma benefit.

Median greeks:

```

VEV_4000 delta 0.9970, gamma 0.000017
VEV_5000 delta 0.9286, gamma 0.000796
VEV_5100 delta 0.8174, gamma 0.001619
VEV_5200 delta 0.6076, gamma 0.002261
VEV_5300 delta 0.3794, gamma 0.002192
VEV_5400 delta 0.1745, gamma 0.001588
VEV_5500 delta 0.0732, gamma 0.000815

```

Gamma-scalp proxy, 3-day net per 1 option unit:

VEV_5200	+29.18
VEV_5300	+28.92
VEV_5400	+22.48
VEV_5100	+20.29
VEV_5500	+11.07
VEV_5000	+9.50
VEV_6500	-0.52
VEV_4000	-0.12

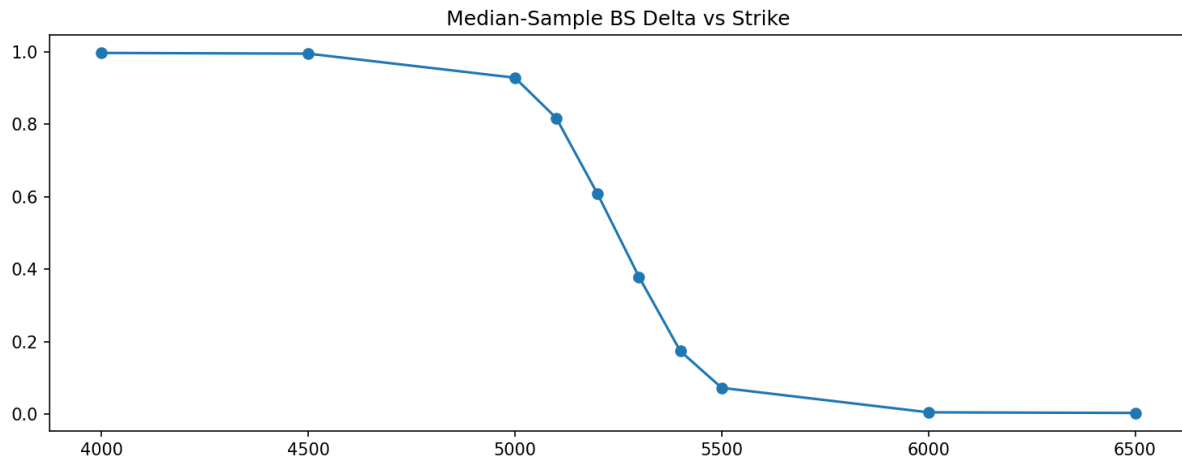


Figure 39: Median-sample Black-Scholes delta by strike.

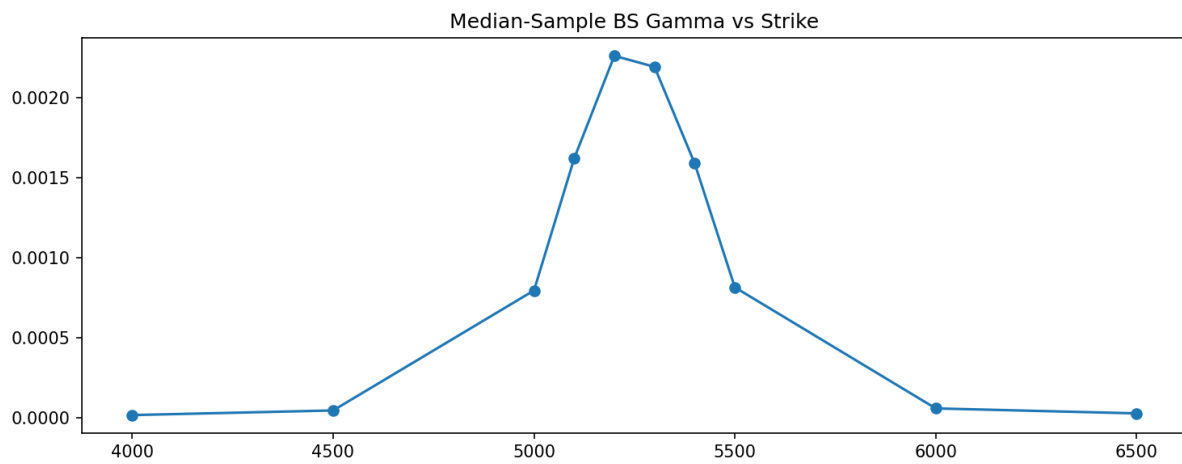


Figure 40: Median-sample Black-Scholes gamma by strike.

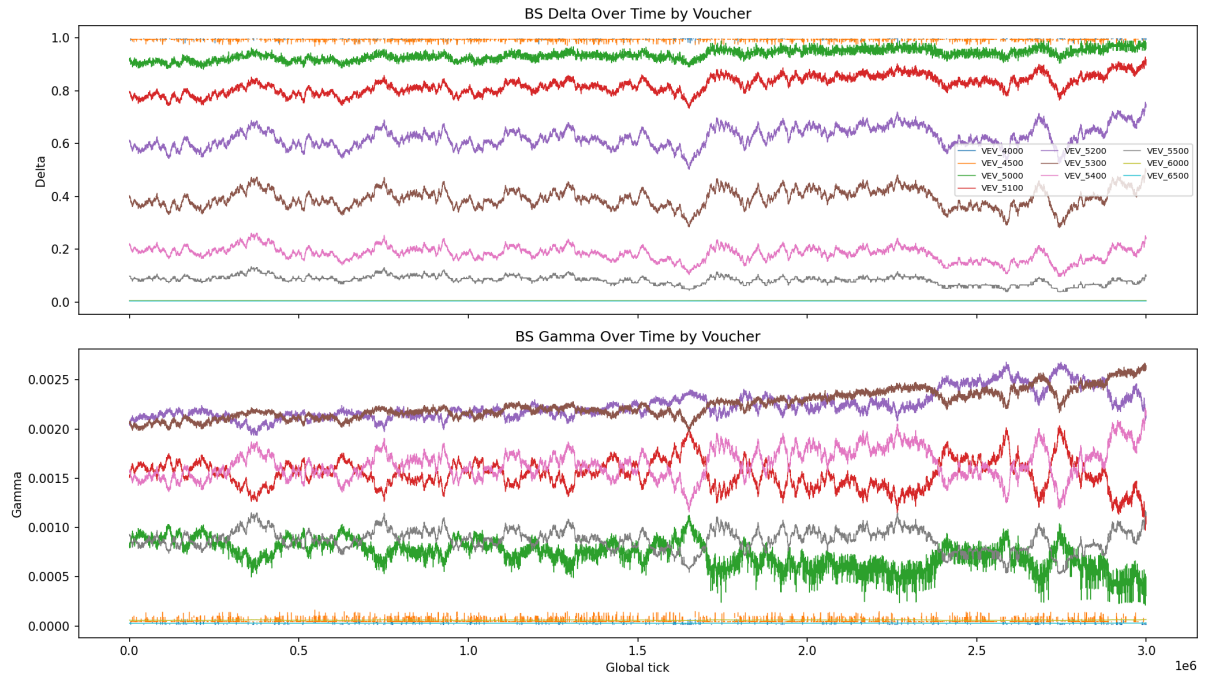


Figure 41: Delta and gamma time series by voucher.



Figure 42: Intrinsic and extrinsic decomposition for selected vouchers.

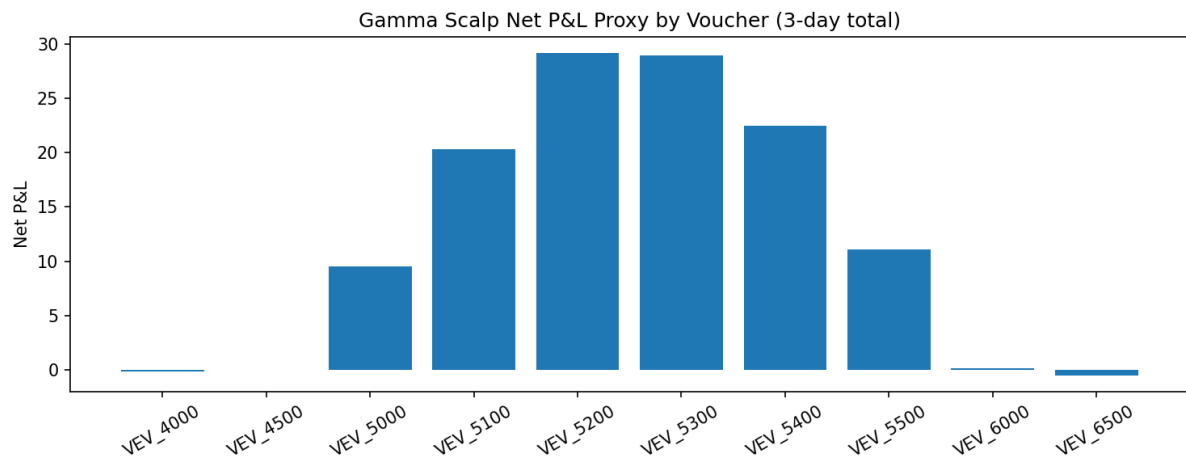


Figure 43: Gamma-scalp PnL proxy summary.

No-Arbitrage Diagnostics

- Lower-bound violations dominate the no-arbitrage table: 11,801 total flags.
- There are no upper-bound violations.

- Lower-bound flags are concentrated in VEV_4500 and VEV_4000.
- There are 9 convexity flags involving VEV_5400, VEV_5500, and VEV_6000.
- Average lower-bound flag lifetimes are short, around 1.1 to 1.5 timestamps.
- Practical strategy must require executable bid/ask edge, persistence, and inventory feasibility before treating these as trades.

Positive no-arbitrage flags:

day	voucher/group	type	count	avg_magnitude	avg_lifetime
0	VEV_4000	lower_bound	939	1.2268	1.097
0	VEV_4500	lower_bound	3045	0.6842	1.485
0	VEV_5400,VEV_5500,VEV_6000	convexity	9	0.5000	1.000
1	VEV_4000	lower_bound	984	1.2175	1.147
1	VEV_4500	lower_bound	3061	0.6900	1.484
2	VEV_4000	lower_bound	1063	1.1834	1.167
2	VEV_4500	lower_bound	2706	0.7308	1.420
2	VEV_5000	lower_bound	3	0.5000	1.000

6 Lead-Lag, Residuals, And Trade-Against-Fair-Value

- Lead-lag correlations peak at lag 0 for all vouchers with meaningful price variation.
- Peak correlations are highest for VEV_5100, VEV_5000, and VEV_5200.
- VEV_6000 and VEV_6500 have undefined lead-lag correlations because their mid prices are pinned.
- Delta-hedged residual standard deviation increases as the option becomes more nonlinear and illiquid.
- Delta-hedged residual autocorrelation is negative across all active vouchers, strongest around VEV_5400 and VEV_5500; this suggests short-horizon residual mean reversion rather than momentum.
- Voucher trades versus Black-Scholes fair value show that VEV_5200 and higher active strikes usually trade below model fair value, while the few valid VEV_4000, VEV_5000, and VEV_5100 trades are above fair value.
- Maker/taker fill breakdown shows aggressive selling pressure in VEV_5200 through VEV_6500: almost all prints occur at the bid.

Lead-lag summary:

voucher	best_lag	peak_corr	residual_std	residual_acf1
VEV_5000	0	0.7524	0.00371	-0.1045
VEV_5100	0	0.7627	0.00502	-0.1025
VEV_5200	0	0.7193	0.00713	-0.1382
VEV_5300	0	0.6123	0.01081	-0.2217
VEV_5400	0	0.5252	0.01770	-0.2508
VEV_5500	0	0.3338	0.02855	-0.2425

Trade price minus BS FV:

VEV_4000	+1.446, but only 46 valid-IV trades.
VEV_5200	-0.917
VEV_5300	-0.963
VEV_5400	-0.629
VEV_5500	-0.547
VEV_6000	-0.500
VEV_6500	-0.500

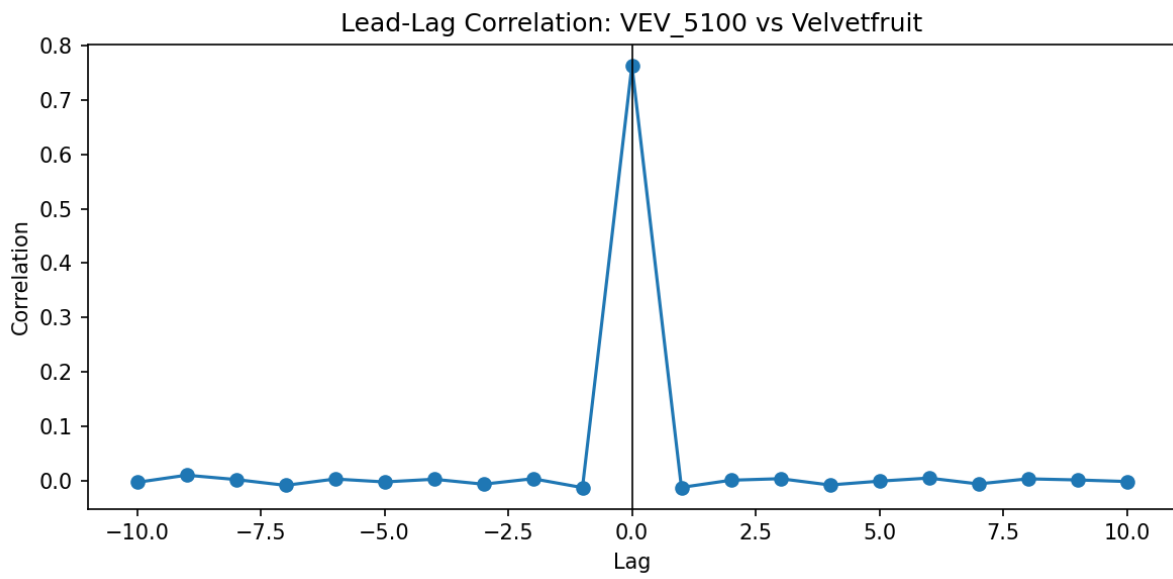


Figure 44: Lead-lag correlation for VEV 5100.

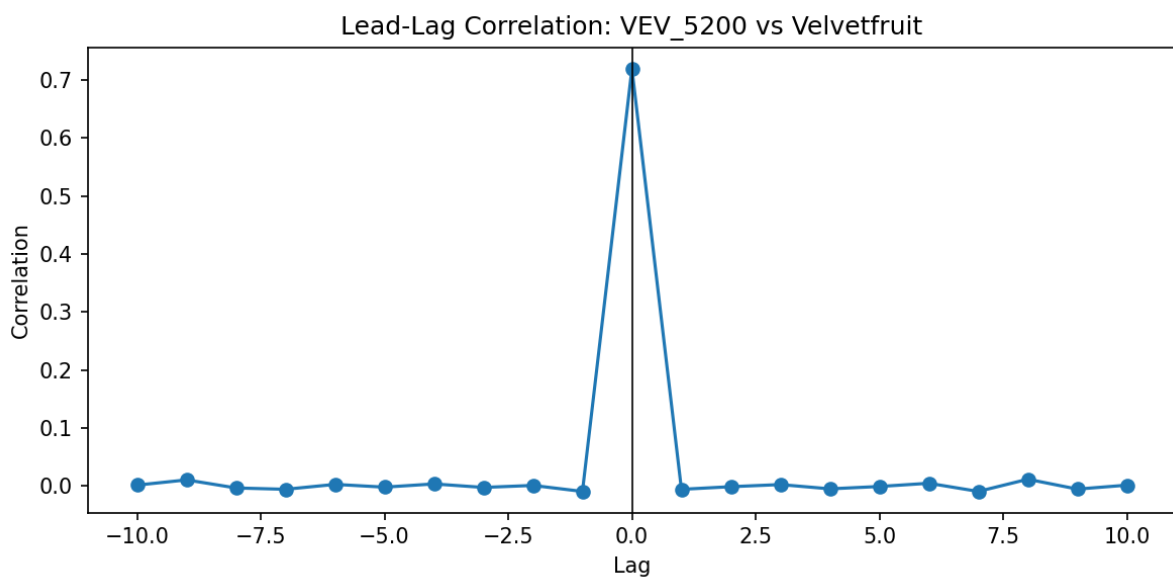


Figure 45: Lead-lag correlation for VEV 5200.

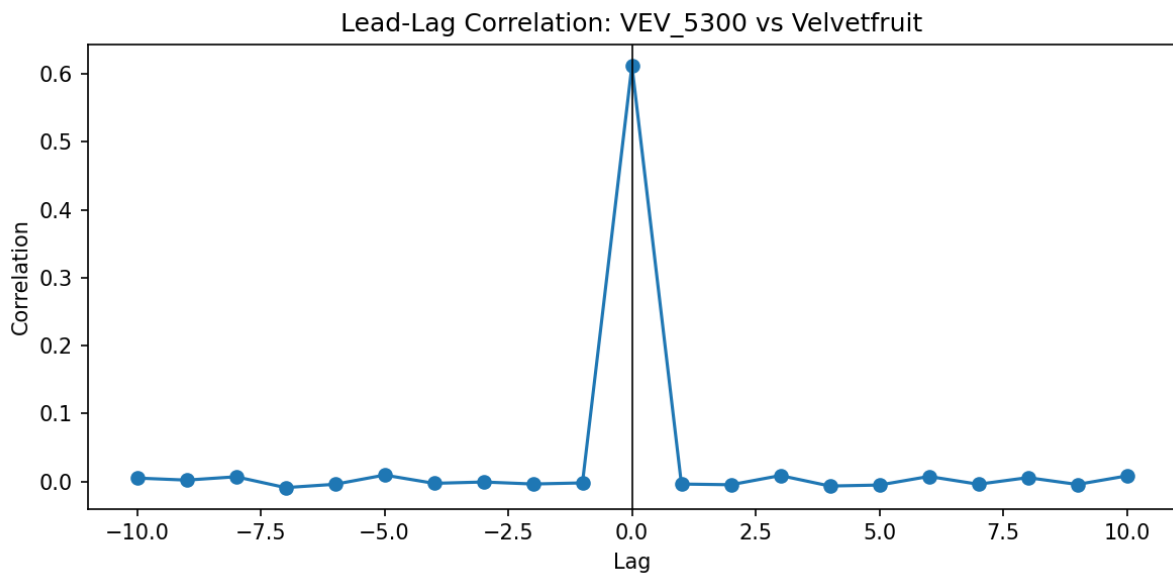


Figure 46: Lead-lag correlation for VEV 5300.

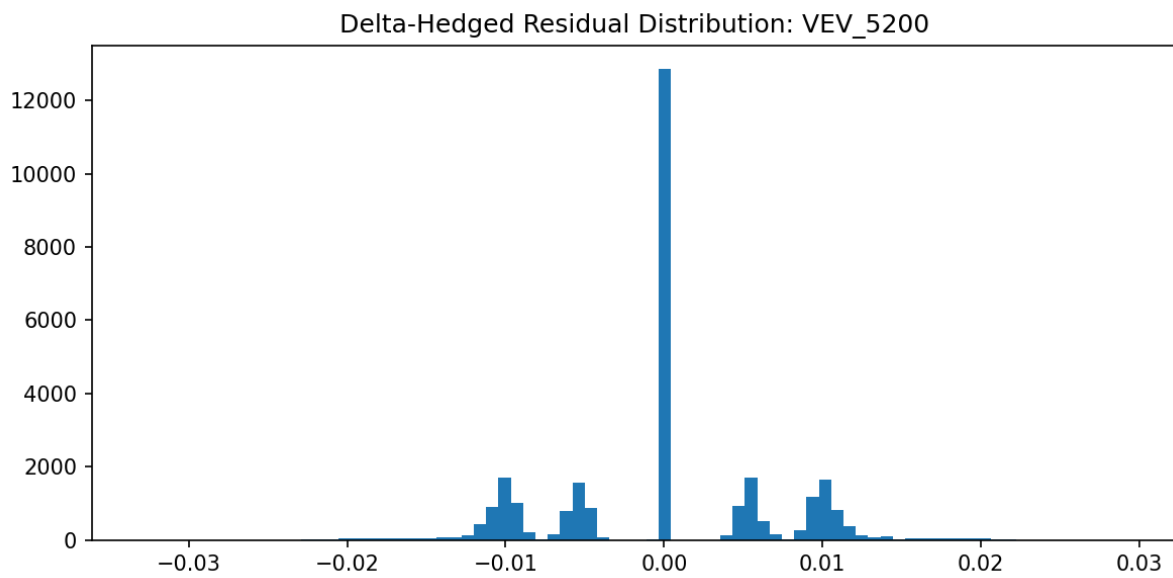


Figure 47: Delta-hedged residual distribution for VEV 5200.

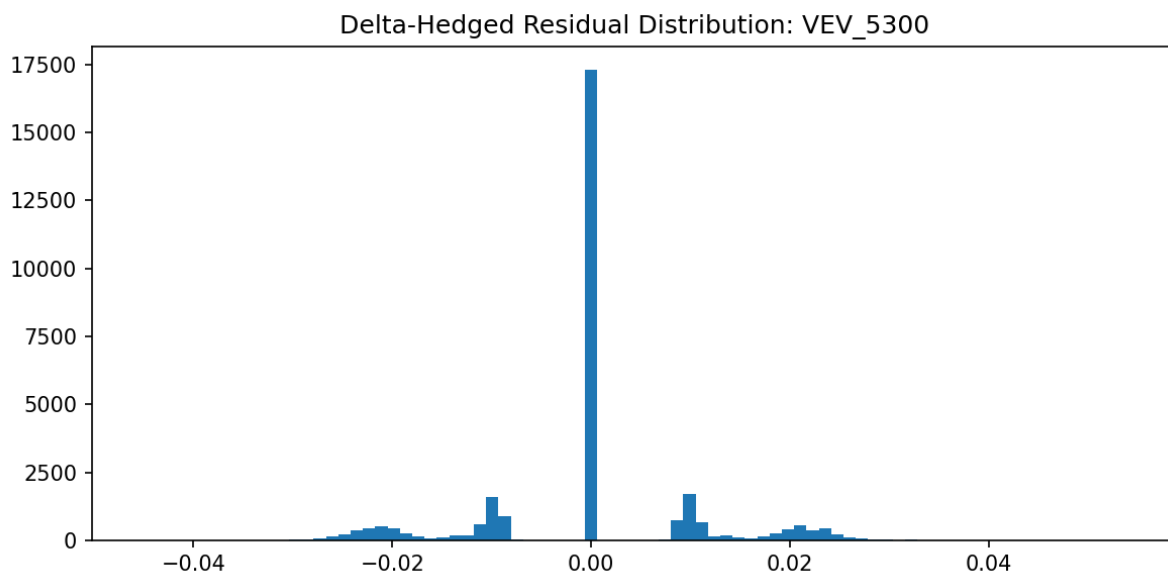


Figure 48: Delta-hedged residual distribution for VEV 5300.

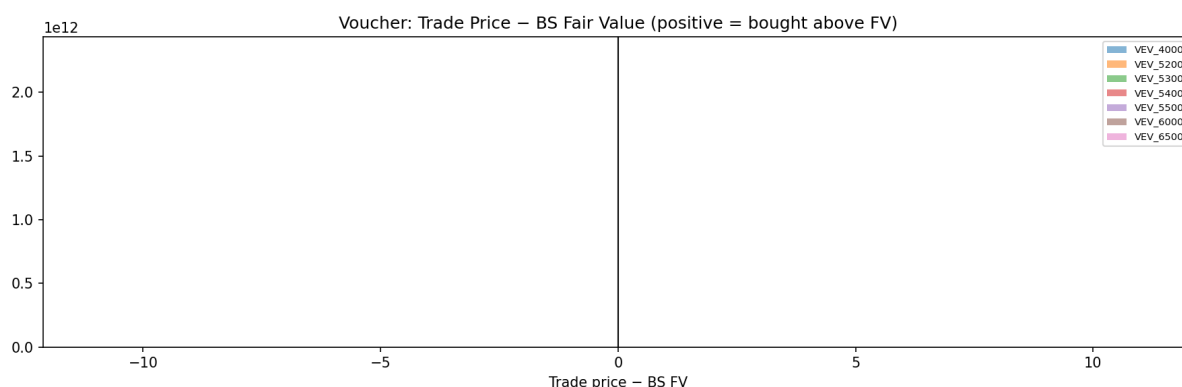


Figure 49: Voucher trade price minus Black-Scholes fair value.

7 Consistency Check Between The Two EDAs

- Target product counts are consistent. The notebook's day-by-day trade counts match the longer EDA's `data_integrity.csv` and `voucher_trade_sizes.csv`.
- Data sources are effectively equivalent for target products: the notebook reads combined files, while the long EDA reads per-day files.
- The main denominator inconsistency is the notebook's product-frequency section: it prints frequencies after filtering out `HYDROGEL_PACK`, while the full dataset has 12 products and 360,000 rows.
- Missing depth is handled differently. The notebook converts missing volumes to zero before computing depth and anomaly states; the long EDA preserves missing price levels and explicitly measures deeper-book availability.
- The notebook is a structural/microstructure EDA, while the long EDA is an option-complex EDA. There is no conflict on IV, greeks, no-arbitrage, or gamma findings because the notebook does not compute those quantities.

- The notebook’s depletion result for `VEV_6500` is consistent with the long EDA’s finding that deep OTM vouchers trade at bid 0 with a bid/ask of 0/1.
- The notebook’s advanced metric `deep_vamp` is useful descriptively, but it should not be compared directly to the longer EDA’s Black-Scholes fair values or delta-hedged residuals.

8 Code Audit Notes

- No fatal syntax error was found in `ve_vev_eda.py`; the file parses successfully.
- A full runtime check using `--output-dir /tmp/vev_eda_check` completed successfully without overwriting the existing output directory.
- The runtime produced a KPSS interpolation warning. This is not fatal; it means the KPSS statistic is outside the tabulated p-value range.
- The requested path `phase2/round3/algo/analysis/ve_vev_eda.py` does not exist in the workspace; the actual script is `phase2/round3/algo/ve_vev_eda.py`.
- `ensure_output()` deletes all existing files in the selected charts and tables directories. This is intended for regeneration but dangerous if pointed at a non-temporary directory.
- The scalar `bs_greeks()` divides vega by 100, while `bs_greeks_vec()` does not. As a result, `voucher_greeks.csv` and `voucher_greeks_timeseries.csv` use different vega units.
- ACF, variance-ratio, Hurst, and stationarity diagnostics are partly computed over the pooled three-day series, so day-boundary jumps can influence the tests.
- The regime label "trending" conflicts with variance ratios below 1.0 and negative return autocorrelation. This should be interpreted as "price level persistent with local mean-reverting ticks", not as simple tradable momentum.
- The IV term structure plot is a historical cross-day proxy, not a true same-time term structure, because TTE changes with day and the underlying path changes too.
- Quote-change rates count the first row of each day as a change because `diff() != 0` treats the initial NaN as changed; the bias is tiny but present.
- `velvetfruit_bot_volume_stats.csv` writes a MultiIndex aggregation without flattening headers, producing an awkward first row containing `mean/std/median`.
- Section numbering in printed runtime messages says 1/6 through 5/6, then 6/7 and 7/7; this is cosmetic.

9 Initial Trading Implications

- Treat `VELVETFRUIT_EXTRACT` as the hedge instrument. Its spread and depth are stable enough for controlled delta hedging, but crossing costs about 2.5 ticks versus mid.
- The largest systematic signal is IV below RV. The natural hypothesis is to own gamma near ATM and delta-hedge, especially in `VEV_5200`, `VEV_5300`, `VEV_5400`, and `VEV_5100`.
- Because voucher spreads and fills are sparse, the strategy should be passive-first in vouchers and selective-taker in the underlying when hedge error becomes too large.
- Deep ITM vouchers are mostly intrinsic value and lower-bound flags are not automatically tradable after spreads.

- Deep OTM floor vouchers should be handled as special discrete-state instruments, not continuous Black-Scholes options.
- A residual mean-reversion overlay can be tested: quote options when delta-hedged residuals are extreme relative to local bands, then hedge with Velvetfruit.
- Any production strategy should combine surface fair value, IV/RV regime, expected fill probability, inventory Greeks, and spread-adjusted expected PnL.

10 100 Research Questions For Further Analysis

- Q1. How stable is the IV-minus-RV gap if RV is computed with windows other than 500 ticks?
- Q2. Does the IV-minus-RV gap predict future delta-hedged option PnL after spread and hedge costs?
- Q3. Which voucher has the highest net expected gamma-scalp PnL after realistic fills, not midpoint marks?
- Q4. Does the gap between ATM IV and RV mean-revert intraday?
- Q5. Does the ATM IV EMA band signal produce profitable entry and exit points?
- Q6. Are VEV_5200 and VEV_5300 always the best ATM proxies, or does the closest strike switch in predictive ways?
- Q7. How much of the gamma-scalp proxy survives if the underlying hedge is executed at bid/ask instead of mid?
- Q8. What hedge threshold minimizes total error plus transaction costs for a long-gamma option book?
- Q9. How often can we obtain voucher fills passively at favorable model prices?
- Q10. Which side of each voucher book receives more aggressive flow conditional on time of day?
- Q11. Does underlying buy-flow toxicity predict voucher mid moves after controlling for delta?
- Q12. Does underlying order-book imbalance become predictive only in high-volume or high-volatility regimes?
- Q13. Is the underlying lag-1 return autocorrelation tradable after spread costs?
- Q14. Are underlying price moves larger after clustered trade-arrival buckets?
- Q15. Does the intraday buy fraction predict the next 1000-timestamp underlying return?
- Q16. Are underlying spreads wider before volatility bursts, or merely contemporaneous with RV?
- Q17. Does total depth predict near-future volatility or trade direction?
- Q18. Do quote offsets from mid change before large underlying moves?
- Q19. Is the bid/ask quote-change rate informative about the next mid-price move?
- Q20. Do outside-top-book underlying trades have special predictive content?
- Q21. Can VELVETFRUIT_EXTRACT be profitably market-made using only mid reversion and stable spread?
- Q22. How does optimal underlying hedge size vary with option gamma and current underlying spread?

- Q23. Does buying at the underlying bid after sell pressure outperform passive random quoting?
- Q24. Are day 1 and day 2 positive drifts persistent enough to justify directional delta?
- Q25. Does the underlying show regime-switching between drift, range, and shock states?
- Q26. Which voucher's delta-hedged residual has the strongest post-cost mean reversion?
- Q27. Are residual bands stable by day, or must they be re-estimated intraday?
- Q28. Do residual shocks in one voucher predict residual shocks in nearby strikes?
- Q29. Does the voucher complex move as a low-dimensional IV surface factor model?
- Q30. Can PCA factors of IV surface predict future option returns better than per-strike IV levels?
- Q31. Are no-arbitrage lower-bound flags executable at bid/ask, or only midpoint artifacts?
- Q32. How long must a no-arbitrage flag persist before it is worth crossing a spread?
- Q33. Are convexity violations around `VEV_5400/VEV_5500/VEV_6000` repeatable outside day 0?
- Q34. Can monotonic and convex smoothing of option mids improve fair-value quotes?
- Q35. Does a constrained IV smile outperform raw mid IV for predicting future trades?
- Q36. Do deep ITM vouchers behave more like stock-plus-cash than options?
- Q37. Is `VEV_4000` useful as a synthetic underlying hedge when the underlying spread is unfavorable?
- Q38. Can a box or vertical-spread portfolio lock in edge after accounting for all bid/ask spreads?
- Q39. Does the option complex imply a forward price that differs from Velvetfruit spot?
- Q40. Are integer-price lower-bound violations caused by rounding rules rather than mispricing?
- Q41. Is there profitable edge in posting bid 0 for `VEV_6000` and `VEV_6500`?
- Q42. Is there profitable edge in posting ask 1 for floor-price OTM vouchers, or is tail risk too high?
- Q43. How should tail risk be modeled for short `VEV_6000/VEV_6500` positions?
- Q44. Do floor-price vouchers ever move away from 0/1 under different live paths?
- Q45. Is the empirical IV of floor-price vouchers meaningful, or should they be modeled as binary lottery tickets?
- Q46. How does TTE decline from 6 historical days to 5 live days change the ATM fair value?
- Q47. Can we extrapolate IV surfaces safely from TTE 8/7/6 to live TTE 5?
- Q48. Does theta decay estimated from historical day transitions match Black-Scholes theta?
- Q49. Is the observed drop in extrinsic value across days mostly theta or underlying-path effect?
- Q50. Which strikes have the most stable extrinsic value after conditioning on moneyness?
- Q51. Are voucher trade sizes informative about informed flow?
- Q52. Does a large voucher trade at bid predict lower future IV?
- Q53. Does a voucher trade at ask predict higher future IV or just temporary demand?
- Q54. How much model edge is needed before crossing a voucher spread is profitable?

- Q55. What passive quote distance maximizes expected fill-adjusted PnL for each strike?
- Q56. Can fill probability be estimated from quote position, spread, and recent trade direction?
- Q57. Are voucher books dominated by stable bots with predictable posted volumes?
- Q58. Does bid volume or ask volume asymmetry in vouchers predict the next voucher trade side?
- Q59. Does voucher quote-change rate decline monotonically with OTM distance because prices are pinned?
- Q60. Can stable volume patterns identify which bot is quoting each strike?
- Q61. Are the buyer/seller columns truly empty in all live-style historical data?
- Q62. How sensitive are Black-Scholes fair values to using calendar-day versus trading-day annualization?
- Q63. Should volatility be annualized using 365 days, 7-day option expiry, or Prosperity-specific tick time?
- Q64. Does realized volatility measured on log returns overstate volatility because of bid/ask bounce?
- Q65. How does RV change after filtering zero changes or using mid-price smoothing?
- Q66. Does using microprice instead of mid improve underlying volatility and delta estimates?
- Q67. Is deep VAMP more predictive of future underlying moves than top-level imbalance?
- Q68. Does DOM-mid deviation predict future reversion to top-of-book mid?
- Q69. Can underlying fair value be improved with Kalman filtering of noisy mid prices?
- Q70. How different are option deltas if fair spot is filtered rather than raw mid?
- Q71. Does delta hedging with filtered spot reduce residual variance?
- Q72. Does a smile-fitted IV produce lower delta-hedged residual variance than raw per-strike IV?
- Q73. Which residual definition best predicts actual trade PnL: price residual, return residual, or dollar hedge error?
- Q74. Can residual autocorrelation be exploited with passive quotes only?
- Q75. Are residuals heteroskedastic by strike and time to expiry?
- Q76. Does residual volatility rise near day boundaries or late-session windows?
- Q77. Is there a consistent intraday seasonality in ATM IV?
- Q78. Does the IV smile steepen or flatten as underlying spot moves?
- Q79. Is skew/smile curvature predictive of future spot direction?
- Q80. Can a straddle or strangle portfolio isolate the IV-vs-RV edge better than single options?
- Q81. Which option portfolio minimizes delta while maximizing gamma under position limits?
- Q82. What is the optimal allocation of the 300-lot voucher limits across strikes?
- Q83. What is the optimal VELVETFRUIT_EXTRACT hedge inventory under a portfolio of voucher positions?

- Q84. How should inventory risk be penalized near the end of the simulation?
- Q85. Does forced mark-to-market at final timestamp favor closing or holding option inventory?
- Q86. What worst-case underlying move breaks a short-OTM voucher strategy?
- Q87. What worst-case spread widening breaks a long-gamma strategy?
- Q88. Are model signals robust if one historical day is held out?
- Q89. Which features survive walk-forward validation from day 0 to day 1 to day 2?
- Q90. Does a live TTE=5 strategy trained on TTE=6 require parameter shrinkage?
- Q91. How should we combine directional underlying alpha with option gamma alpha?
- Q92. Can we synthesize a better underlying exposure using deep ITM vouchers plus cash?
- Q93. What are the PnL effects of integer order prices and tick-size rounding?
- Q94. How often does rounding flip a theoretical edge from positive to negative?
- Q95. What is the best quoting grid for each voucher after fair-value rounding?
- Q96. Which signals remain profitable after adverse-selection penalties from fills?
- Q97. Can a joint simulator replay historical order books and test passive fills realistically?
- Q98. Which EDA signal is most likely to overfit only three historical days?
- Q99. What minimal strategy captures most of the IV/RV edge with the least operational risk?
- Q100. What live-regime guardrail should respond to RV collapse, IV spikes, or hedge PnL deviation?

11 30 Candidate Models And Validation Statistics

- M1. **Random walk with drift for underlying mid.** Validate with out-of-sample log-likelihood, drift t-statistics, ADF/KPSS on residual increments, and variance-ratio tests.
- M2. **ARMA/ARIMA on underlying returns.** Validate with AIC/BIC, Ljung-Box residual tests, residual ACF, walk-forward RMSE, and directional hit rate.
- M3. **ARFIMA or fractional-noise model for price persistence.** Validate with Hurst consistency, Whittle likelihood, residual long-memory tests, and out-of-sample forecast loss.
- M4. **Ornstein-Uhlenbeck model for microstructure return reversion.** Validate with half-life stability, residual normality, likelihood ratio versus random walk, and post-cost mean-reversion PnL.
- M5. **Regime-switching AR model for drift/range/shock states.** Validate with hidden-state persistence, out-of-sample log score, transition stability by day, and strategy PnL by inferred regime.
- M6. **GARCH(1,1) for underlying one-tick returns.** Validate with QLIKE volatility loss, ARCH-LM residual tests, standardized residual kurtosis, and volatility forecast MSE.
- M7. **EGARCH or GJR-GARCH for asymmetric volatility.** Validate with leverage-term significance, AIC/BIC improvement over GARCH, VaR exception counts, and residual diagnostics.

- M8. **HAR-RV model using RV100/RV500/RV2000.** Validate with out-of-sample RV forecast RMSE/MAE, QLIKE, Diebold-Mariano tests versus EWMA, and coefficient stability.
- M9. **EWMA/RiskMetrics volatility model.** Validate lambda via walk-forward likelihood, RV forecast error, and sensitivity of option fair values to lambda.
- M10. **Kalman local-level model for filtered underlying fair value.** Validate with one-step prediction error, filtered-versus-raw hedge residual variance, and likelihood.
- M11. **Order-book imbalance linear model for forward returns.** Validate with t-statistics, adjusted R^2 , robust standard errors, sign hit rate, and post-spread PnL.
- M12. **Nonlinear tree or spline model for imbalance and depth features.** Validate with cross-validated R^2 , permutation importance, calibration by feature bucket, and walk-forward PnL.
- M13. **Ordered logit/probit for next tick move direction.** Validate with log loss, Brier score, confusion matrix, calibration curves, and economic value after costs.
- M14. **Poisson or negative-binomial model for trade arrivals.** Validate with dispersion index, likelihood ratio test, Pearson residuals, and intraday bucket calibration.
- M15. **Hawkes process for clustered trades.** Validate with time-rescaling KS test, branching ratio stability, likelihood improvement over Poisson, and burst-period PnL attribution.
- M16. **Autoregressive conditional duration model for interarrival times.** Validate with standardized duration diagnostics, AIC/BIC, and forecasted fill-risk calibration.
- M17. **Black-Scholes fair-value model with empirical IV.** Validate with trade-minus-fair residuals, delta-hedged residual variance, IV status coverage, and post-cost option PnL.
- M18. **Constrained quadratic IV smile by timestamp.** Validate with smile R^2 , no-arbitrage violation count, out-of-sample trade-price error, and residual smoothness.
- M19. **SVI-style parametric implied-volatility surface.** Validate with weighted IV RMSE, convexity/no-arbitrage checks, parameter stability, and bid/ask fit rate.
- M20. **SABR-inspired smile approximation.** Validate with strike-weighted IV error, curvature fit around ATM, parameter plausibility, and delta/gamma hedge error.
- M21. **PCA/factor model of IV surface changes.** Validate with explained variance, factor autocorrelation, out-of-sample reconstruction error, and factor-PnL relationship.
- M22. **Bayesian hierarchical IV model by strike and day.** Validate posterior predictive checks, credible-interval coverage, leave-one-day-out likelihood, and shrinkage robustness.
- M23. **Delta-hedged residual AR(1) model by voucher.** Validate residual half-life, residual ACF, z-score calibration, mean-reversion PnL, and turnover-adjusted Sharpe.
- M24. **Gamma-scalping PnL attribution model.** Validate realized $0.5\Gamma dS^2 + \Theta dt$ attribution, hedge-cost drag, net PnL by strike, and sensitivity to hedge frequency.
- M25. **No-arbitrage persistence survival model.** Validate hazard calibration, precision/recall for persistent flags, spread-adjusted opportunity PnL, and false-positive rate.
- M26. **Constrained cross-sectional option-pricing regression.** Validate monotonicity, convexity, bid/ask inclusion rate, residual RMSE, and executable arbitrage count.
- M27. **Voucher fill-probability logistic model.** Validate ROC-AUC, precision at top quote candidates, calibration curves, and expected fill-adjusted PnL.

- M28. **Avellaneda-Stoikov style market-making model with option Greeks.** Validate inventory distribution, spread capture, adverse-selection loss, drawdown, and risk-adjusted PnL.
- M29. **Linear-quadratic inventory control for delta/gamma constrained portfolios.** Validate terminal inventory penalty, hedge turnover, realized PnL distribution, and stress-test loss.
- M30. **Walk-forward ensemble combining IV/RV, residual, flow, and inventory signals.** Validate day-held-out PnL, feature ablation, turnover-adjusted Sharpe, maximum drawdown, and live TTE=5 sensitivity.

12 Appendix: Additional Plot Panels

The following panels include the main plot families generated by the longer EDA. They are deliberately kept as visual appendices so the core research narrative stays readable.

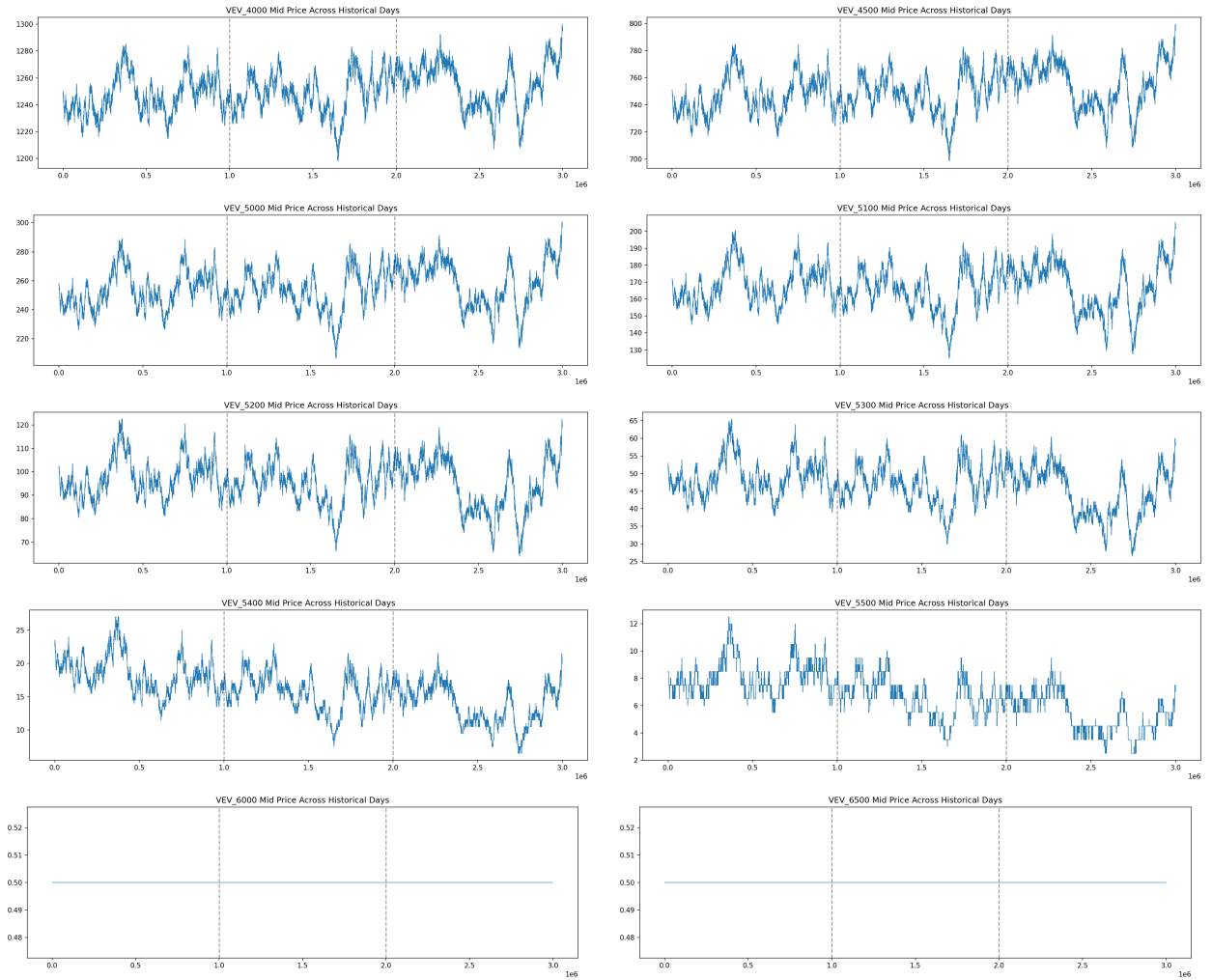


Figure 50: Per-voucher mid-price time series.

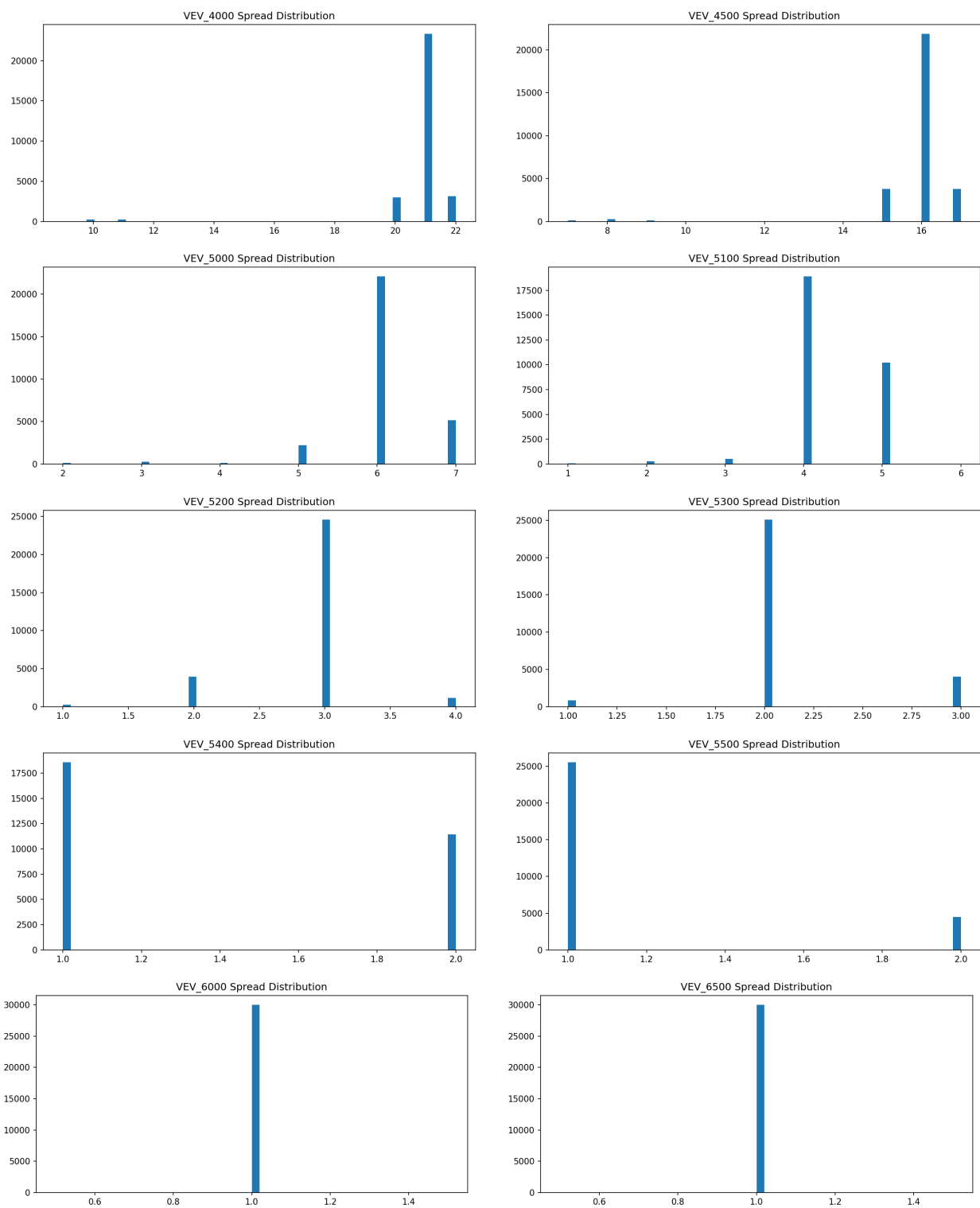


Figure 51: Per-voucher spread distributions.

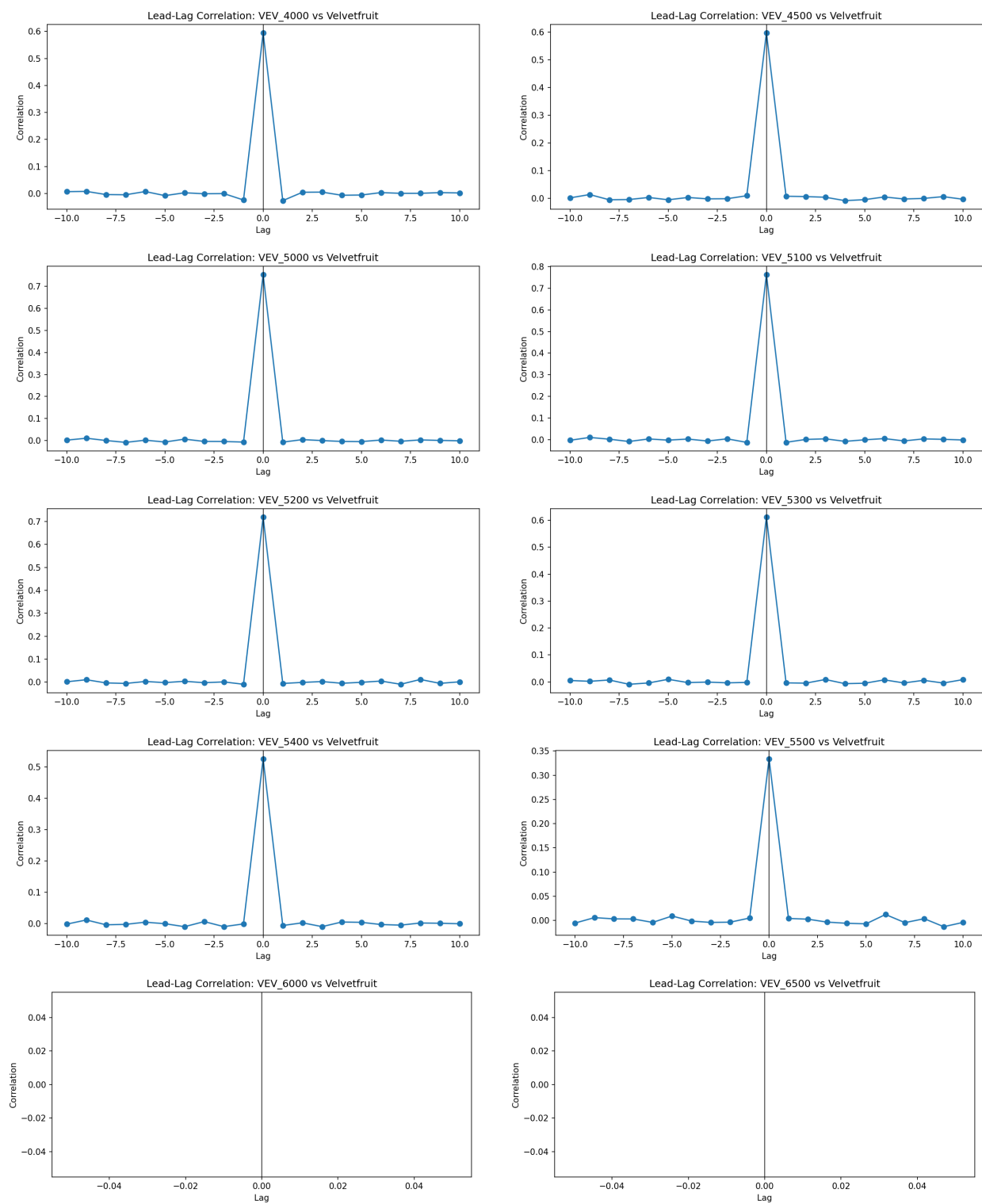


Figure 52: Lead-lag correlation panels by voucher.

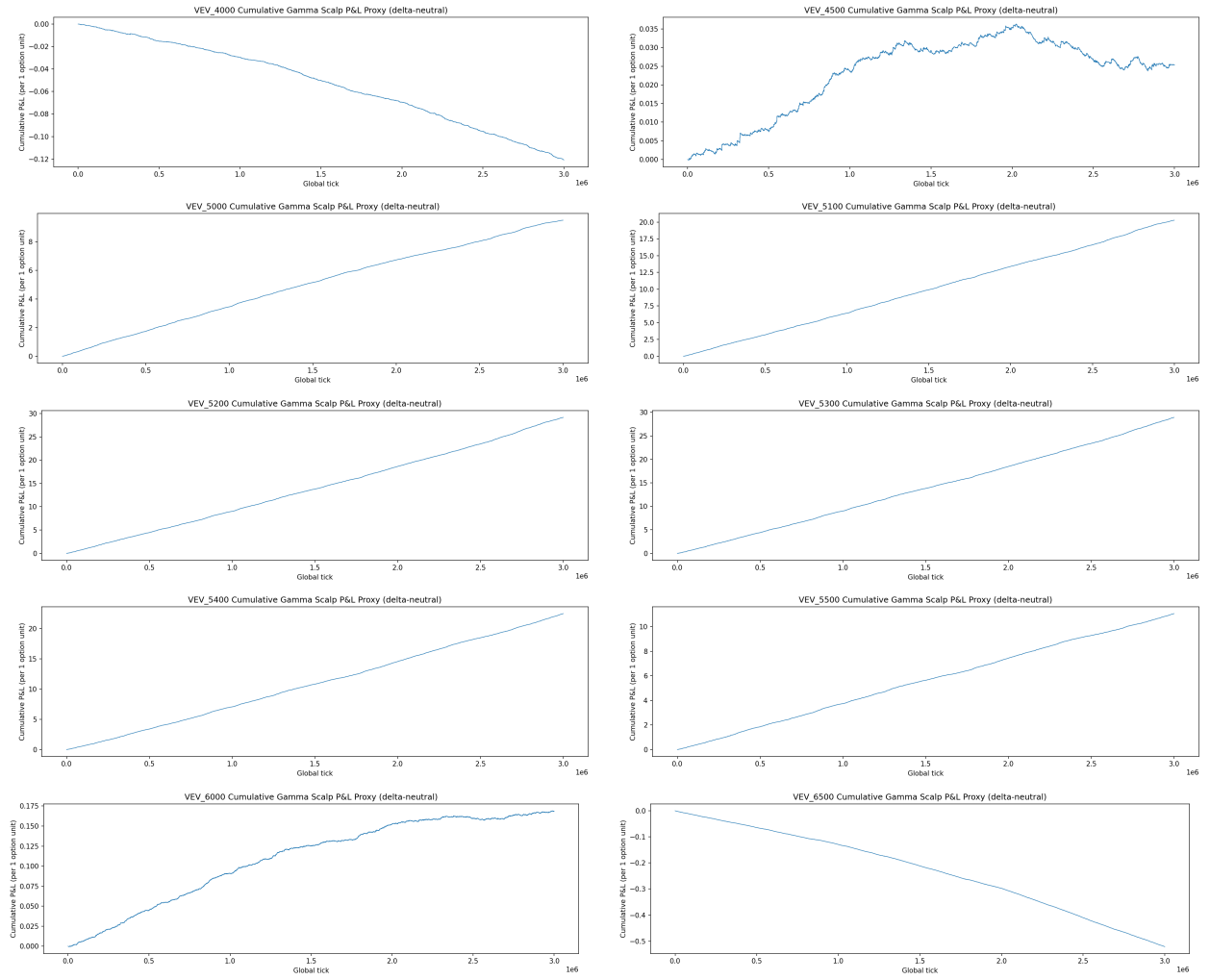


Figure 53: Gamma-scalp proxy time series by voucher.

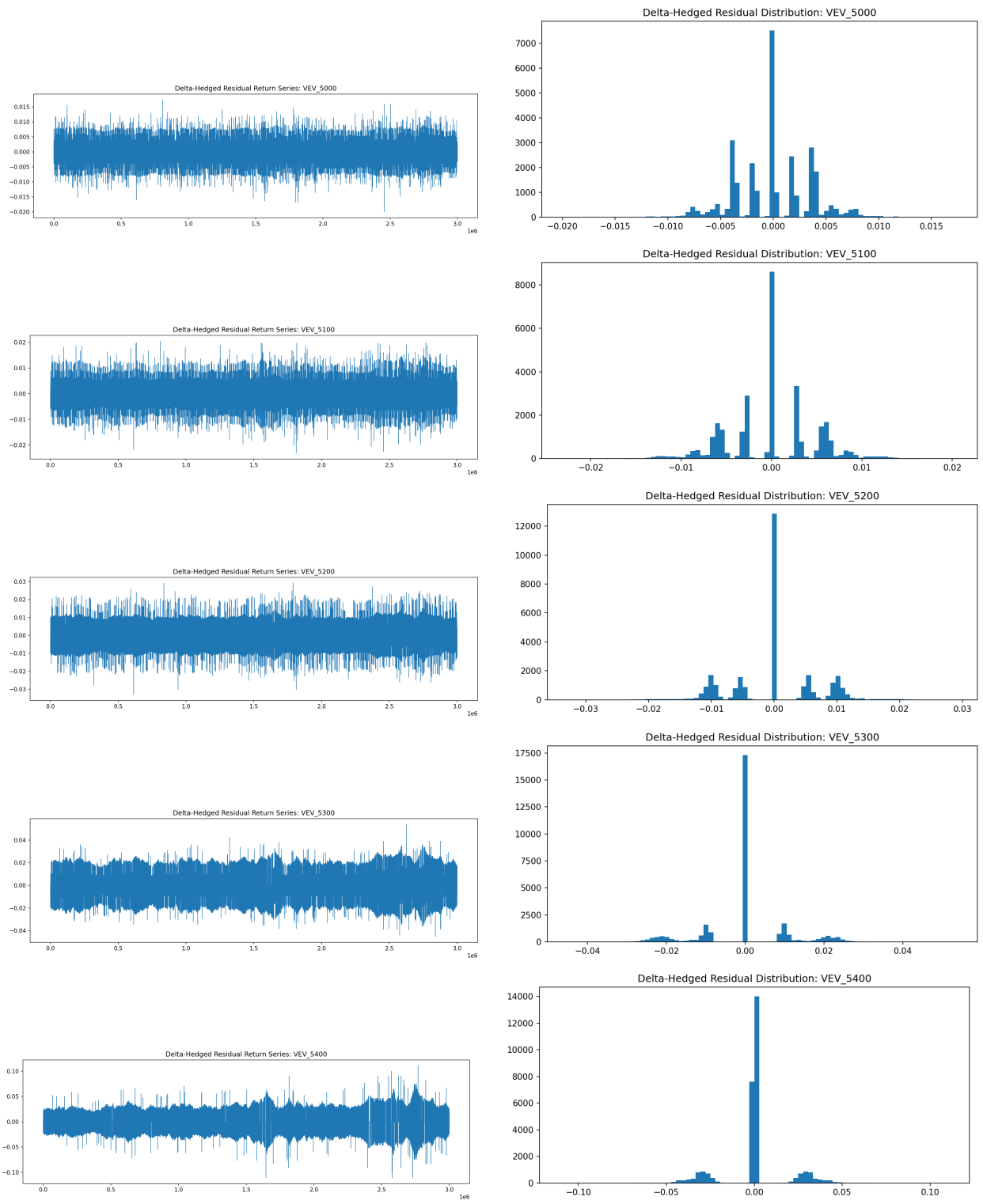


Figure 54: Representative delta-hedged residual series and histograms.